Machine Learning Model for Predicting Syriatel Customer Churn

Project Overview

This project aims to support SyriaTel in addressing customer churn by developing a classification model that predicts whether a customer is likely to stop doing business with the company in the near future. By accurately predicting customer churn, SyriaTel can take early action to address customer dissatisfaction, implement targeted retention measures, and strengthen customer loyalty. Ultimately, this will help reduce churn rates, improve customer satisfaction, and enhance the company's competitive position in the market.

Business Understanding

SyriaTel is a telecommunications company that provides mobile network services in Syria. Recently, the company has experienced rising customer attrition rates, which pose a serious threat to revenue stability and market share. Customer churn is a metric showing when a subscriber or a regular customer cancels his subscription or stops doing business with a company. This challenge highlights the need to better understand the key factors contributing to customer churn. Identifying the drivers of customer defection and developing proactive retention strategies is essential for maintaining profitability and remaining competitive in the telecommunications industry.

Problem Statement:

As SyriaTel navigates a competitive telecommunications landscape, the increasing attrition of customers poses a significant challenge to its growth and profitability. By analyzing the factors contributing to churn, SyriaTel can adopt proactive measures to enhance customer satisfaction and loyalty, ultimately improving retention rates thereby ensuring sustained profitability in a competitive market.

Objectives:

1. Identify Churn Patterns: Analyze customer data to uncover patterns associated with churn.
2. Develop a Customer Churn Prediction Model: Design a binary classification model employing machine learning techniques to estimate the probability of a customer discontinuing services. The model should be trained on past customer data and evaluated with suitable performance metrics such as accuracy, precision, recall, and F1-score.
3. Deliver Practical Recommendations: Provide practical recommendations and insights derived from analysis and model outcomes.

Stakeholder

The stakeholder is the Syriatel company including the business and customer retention team

Data Understanding

The Customer Churn in Telecom's dataset from Kaggle contains information about customer activity and whether or not they canceled their subscription with the Telecom firm. The goal of this dataset is to develop predictive models that can help the telecom business reduce the amount of money lost due to customers who don't stick around for very long.

The dataset contains 3333 entries and 21 columns, including information about the state, account length, area code, phone number, international plan, voice mail plan, number of voice mail messages, total day minutes, total day calls, total day charge, total evening minutes, total evening calls, total evening charge, total night minutes, total night calls, total night charge, total international minutes, total international calls, total international charge, customer service calls and churn.

1. Load data

In [9]:

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import roc_curve, auc
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDispl
ay
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_selection import RFECV
```

In [10]:

```
# Load the dataset
customer_data = pd.read_csv('SyriaTel_Churn_Dataset.csv')
# Display the first few rows of the dataset
customer_data.head()
```

Out[10]:

	state	account length	area code	phone number	international plan	voice mail plan	number vmail messages	total day minutes	total day calls	total day charge	...	total eve calls	total eve charge	total night minutes	total night calls	ch
0	KS	128	415	382-4657	no	yes	25	265.1	110	45.07	...	99	16.78	244.7	91	
1	OH	107	415	371-7191	no	yes	26	161.6	123	27.47	...	103	16.62	254.4	103	
2	NJ	137	415	358-1921	no	no	0	243.4	114	41.38	...	110	10.30	162.6	104	
3	OH	84	408	375-9999	yes	no	0	299.4	71	50.90	...	88	5.26	196.9	89	
4	OK	75	415	330-6626	yes	no	0	166.7	113	28.34	...	122	12.61	186.9	121	

5 rows × 21 columns

In [11]:

```
# Preview the dataset's shape and data types
print(customer_data.shape)
print(customer_data.info())
```

```
(3333, 21)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3333 entries, 0 to 3332
Data columns (total 21 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   state                   3333 non-null   object
 1   account length          3333 non-null   int64
 2   area code               3333 non-null   int64
 3   phone number            3333 non-null   object
 4   international plan      3333 non-null   object
 5   voice mail plan         3333 non-null   object
 6   number vmail messages   3333 non-null   int64
```

```
 7   total day minutes       3333 non-null   float64
 8   total day calls         3333 non-null   int64
 9   total day charge        3333 non-null   float64
 10  total eve minutes       3333 non-null   float64
 11  total eve calls         3333 non-null   int64
 12  total eve charge        3333 non-null   float64
 13  total night minutes     3333 non-null   float64
 14  total night calls       3333 non-null   int64
 15  total night charge      3333 non-null   float64
 16  total intl minutes      3333 non-null   float64
 17  total intl calls        3333 non-null   int64
 18  total intl charge       3333 non-null   float64
 19  customer service calls  3333 non-null   int64
 20  churn                   3333 non-null   bool
dtypes: bool(1), float64(8), int64(8), object(4)
memory usage: 524.2+ KB
None
```

2. Data Cleaning

In [12]:

```
# Check for missing values
print(customer_data.isnull().sum())
```

```
state                     0
account length            0
area code                 0
phone number              0
international plan         0
voice mail plan           0
number vmail messages     0
total day minutes         0
total day calls           0
total day charge          0
total eve minutes         0
total eve calls           0
total eve charge          0
total night minutes       0
total night calls         0
total night charge        0
total intl minutes        0
total intl calls          0
total intl charge         0
customer service calls    0
churn                     0
dtype: int64
```

In [13]:

```
# Check for duplicates
print(customer_data.duplicated().sum())
```

```
0
```

Our dataset has no duplicates or missing values. We can go ahead and do data exploration

3. Exploratory Data analysis

Our target variable is 'Churn', which indicates whether a customer has churned (1) or not (0). We will analyze the distribution of this target variable to understand the balance of our dataset.

In [14]:

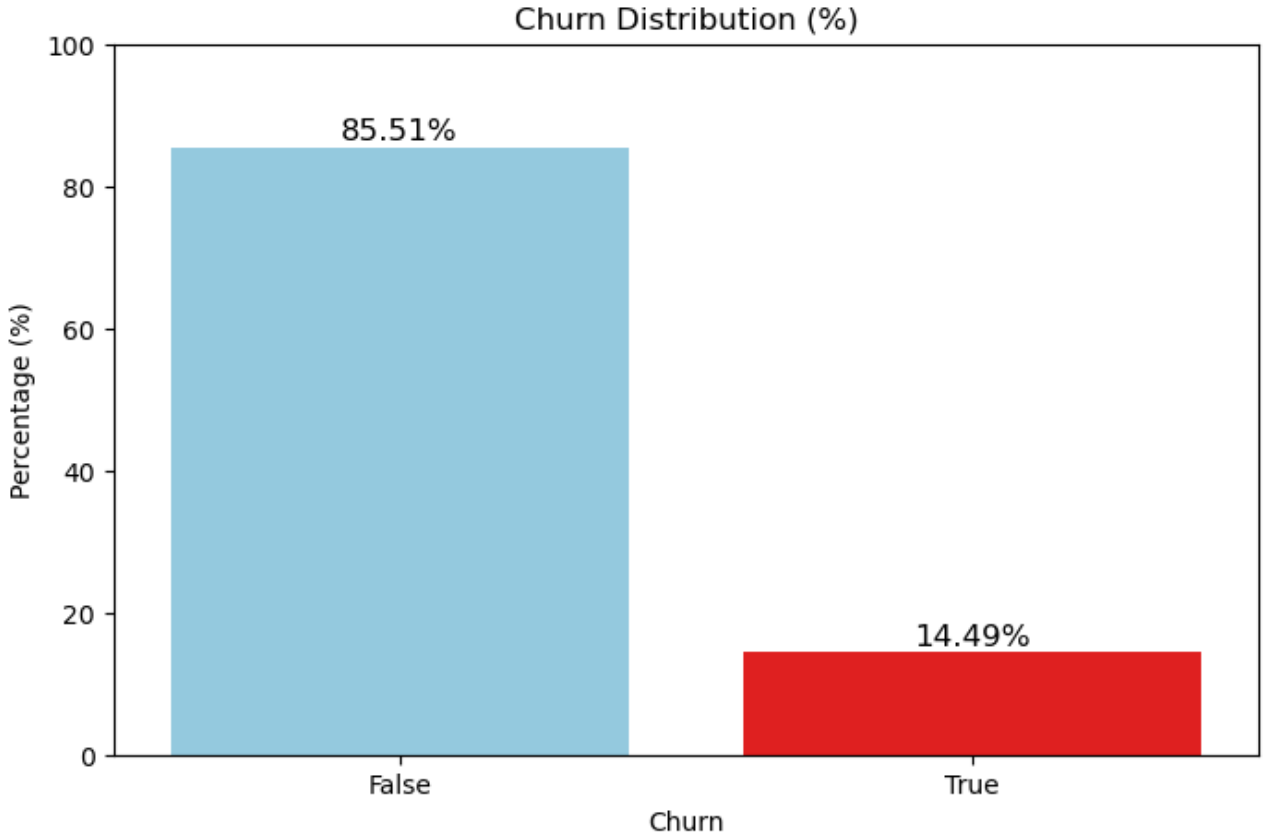
```
# Check target variable distribution
print(customer_data['churn'].value_counts())
```

```
churn
```

```
churn
False    2850
True      483
Name: count, dtype: int64
```

In [15]:

```
# Calculate percentages
churn_percent = (customer_data['churn'].value_counts(normalize=True) * 100).reset_index(
)
churn_percent.columns = ['churn', 'percent']

# Plot bar chart
plt.figure(figsize=(8,5))
ax = sns.barplot(
    data=churn_percent,
    x='churn',
    y='percent',
    hue='churn',
    palette={False: 'skyblue', True: 'red'},
    legend=False
)

# Add percentage labels
for i, v in enumerate(churn_percent['percent']):
    ax.text(i, v + 1, f"{v:.2f}%", ha='center', fontsize=12)

plt.title("Churn Distribution (%)")
plt.xlabel("Churn")
plt.ylabel("Percentage (%)")
plt.ylim(0, 100)
plt.show()
print(f"85.51% of the data belongs to the False class while 14.49% belongs to the true cl
ass, this indicates that the dataset is imbalanced.")
```




```
85.51% of the data belongs to the False class while 14.49% belongs to the true class, thi
s indicates that the dataset is imbalanced.
```

In [16]:

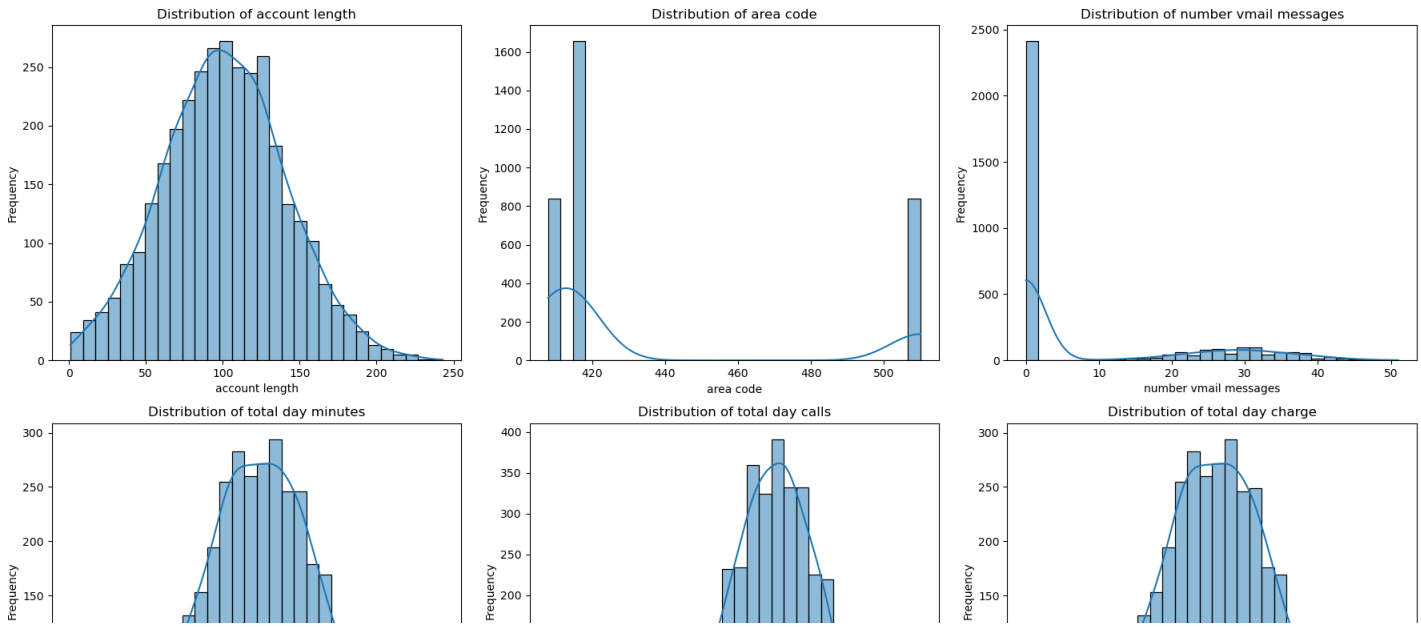
```
# Get summary statistics for numerical features
customer_data.describe()
```

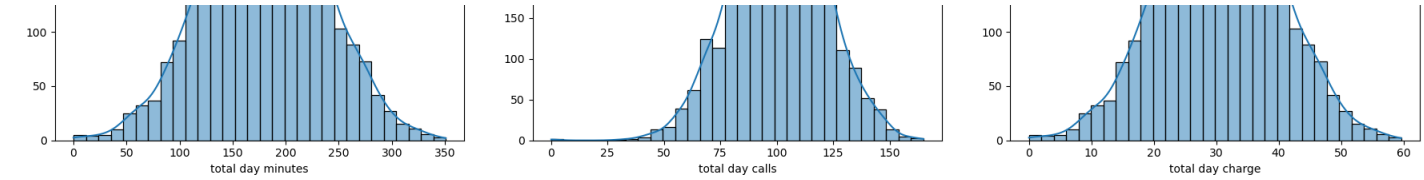
Out[16]:

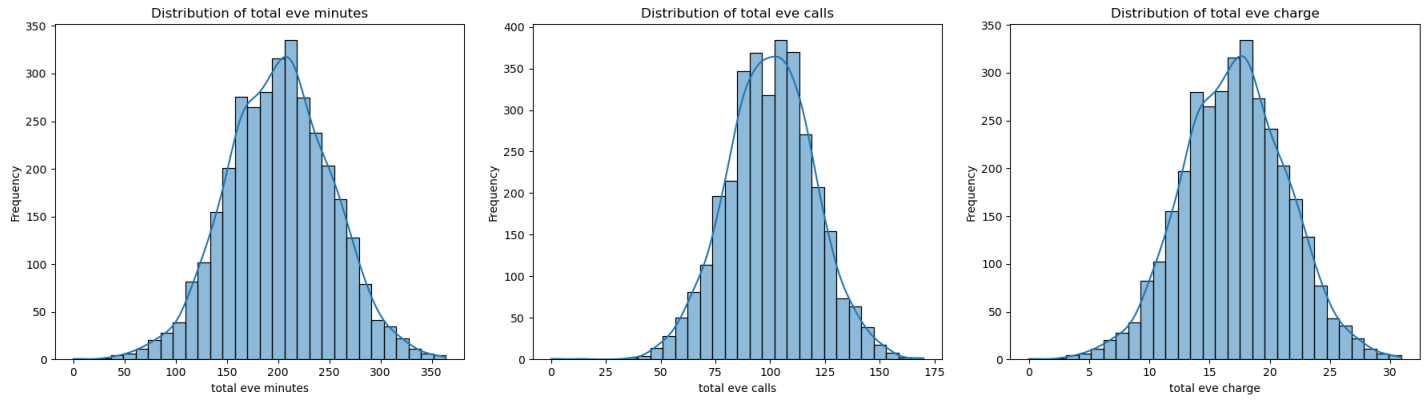
	account length	area code	number vmail messages	total day minutes	total day calls	total day charge	total eve minutes	total eve calls	total eve charge
count	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000	3333.000000
mean	101.064806	437.182418	8.099010	179.775098	100.435644	30.562307	200.980348	100.114311	17.083540
std	39.822106	42.371290	13.688365	54.467389	20.069084	9.259435	50.713844	19.922625	4.310668
min	1.000000	408.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	74.000000	408.000000	0.000000	143.700000	87.000000	24.430000	166.600000	87.000000	14.160000
50%	101.000000	415.000000	0.000000	179.400000	101.000000	30.500000	201.400000	100.000000	17.120000
75%	127.000000	510.000000	20.000000	216.400000	114.000000	36.790000	235.300000	114.000000	20.000000
max	243.000000	510.000000	51.000000	350.800000	165.000000	59.640000	363.700000	170.000000	30.910000

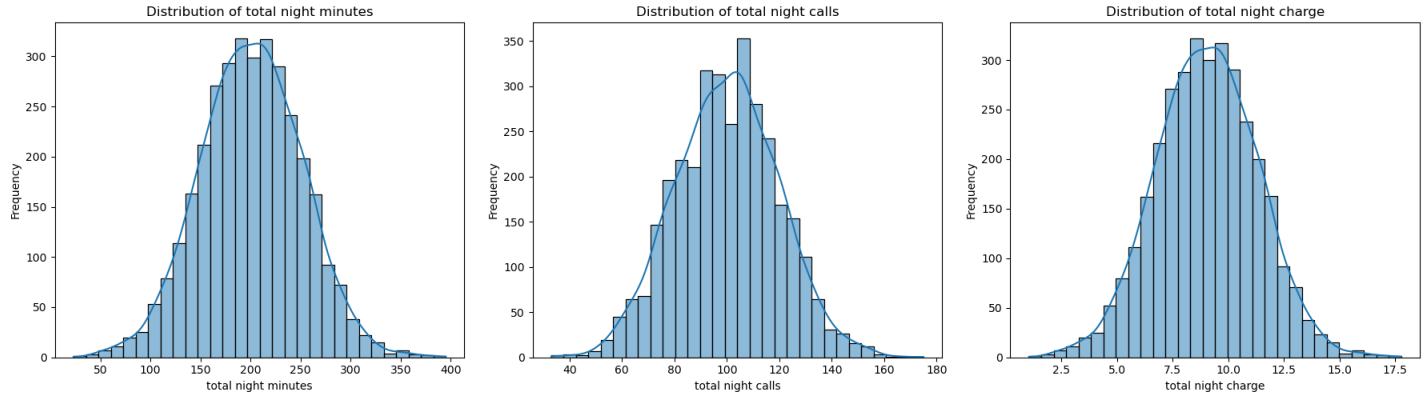
In [17]:

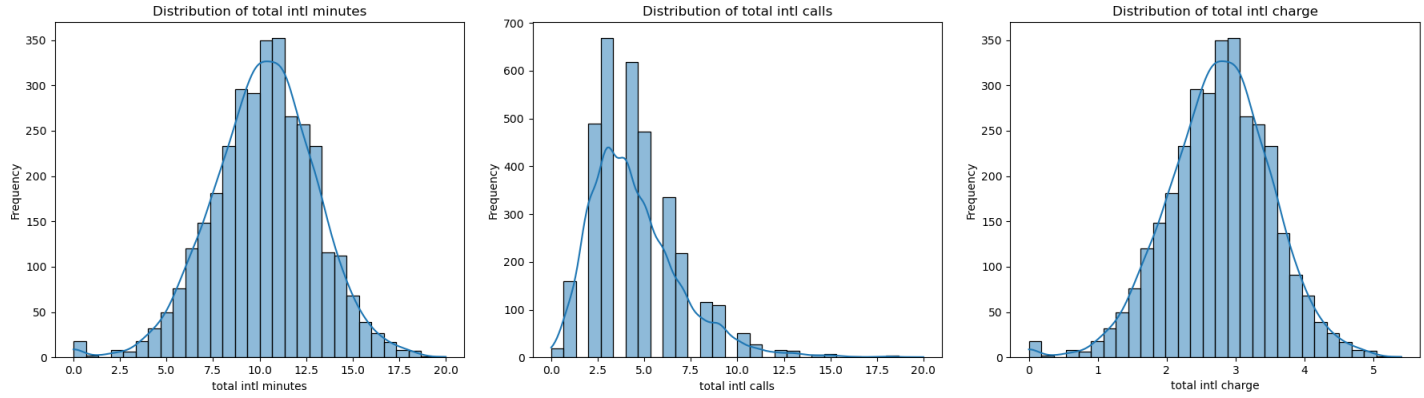
```
# Plotting distributions of numerical features,
# we will use histograms with KDE (Kernel Density Estimate) to
# visualize the distribution of each numerical feature in the dataset.

# Numerical features
import math

numerical_features = customer_data.select_dtypes(include=[np.number]).columns.tolist()

# subplot layout
n_cols = 3
n_rows = math.ceil(len(numerical_features) / n_cols)

fig, axes = plt.subplots(n_rows, n_cols, figsize=(18, 5 * n_rows))
axes = axes.flatten()

# Plot each feature
for i, feature in enumerate(numerical_features):
    sns.histplot(customer_data[feature], kde=True, bins=30, ax=axes[i])
    axes[i].set_title(f"Distribution of {feature}")
    axes[i].set_xlabel(feature)
    axes[i].set_ylabel("Frequency")

# Remove empty subplots
for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```

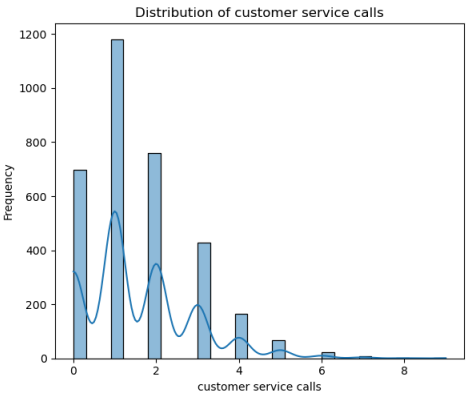












Some features show a right-skewed distribution, while others appears to be more normally distributed.

This information will help us decide on potential transformations or scaling methods for these features during preprocessing.

Bivariate Analysis

Bivariate analysis involves analyzing the relationship between two variables. For this project, I examine the relationship between each feature and the target variable (customer churn) to understand how they are related.

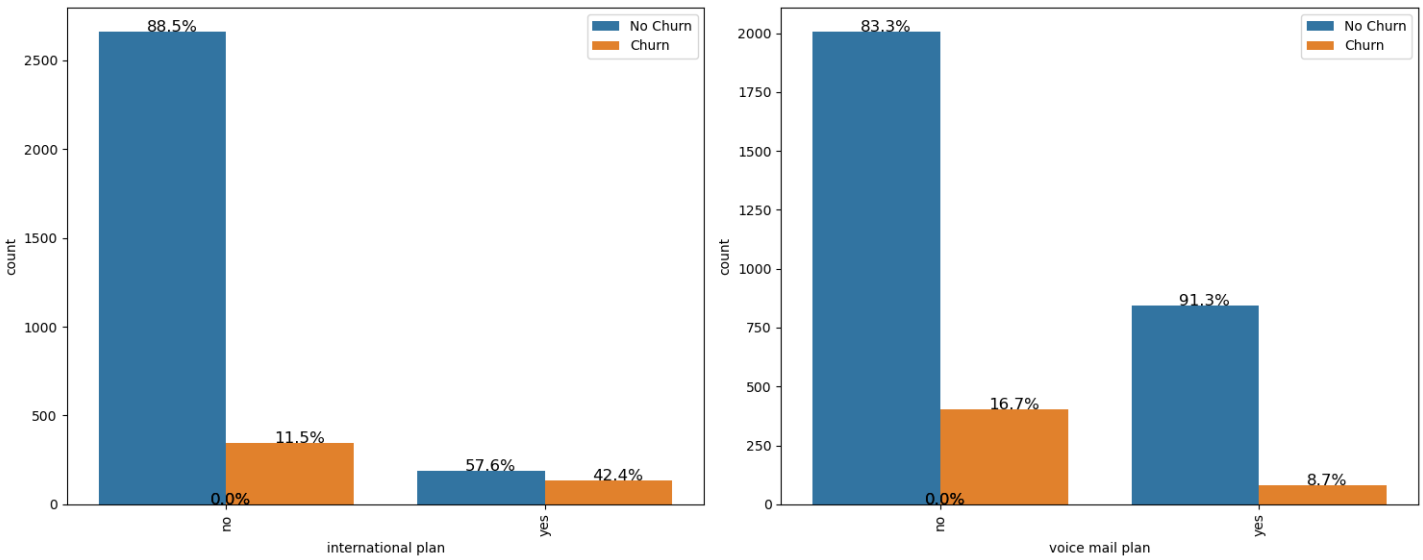
Here, I am doing some analysis of the customer churning in relation to international plan and voice mail plan. I

Here, I am doing some analysis of the customer churning in relation to international plan and voice mail plan. I am trying to understand whether there are correlations between the categorical columns and the customer churning rate.

In [18]:

```
categoric_cols = ['international plan','voice mail plan']

fig, axes = plt.subplots(nrows=1, ncols=len(categoric_cols), figsize=(15, 6))

for i, col in enumerate(categoric_cols):
    ax = sns.countplot(x=col, hue="churn", data=customer_data, order=customer_data[col].
value_counts().iloc[0:15].index, ax=axes[i])
    axes[i].set_xticklabels(axes[i].get_xticklabels(), rotation=90)
    handles, labels = axes[i].get_legend_handles_labels()
    axes[i].legend(handles, ['No Churn', 'Churn'], loc="upper right")

    # Calculate the total number of observations within each group
    totals = customer_data.groupby(col)["churn"].count().values

    # Iterate over the rectangles in the plot
    for j, p in enumerate(ax.patches):
        # Calculate the percentage of observations in each group
        percentage = '{:.1f}%'.format(100 * p.get_height()/totals[j % 2])
        # Add text annotations with the calculated percentages
        x = p.get_x() + p.get_width() / 2 - 0.05
        y = p.get_y() + p.get_height()
        ax.annotate(percentage, (x, y), size=12)

plt.tight_layout()
plt.show()
```

```
C:\Users\Stephen\AppData\Local\Temp\ipykernel_26400\2444976805.py:7: UserWarning: set_tic
klabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or usi
ng a FixedLocator.
  axes[i].set_xticklabels(axes[i].get_xticklabels(), rotation=90)
C:\Users\Stephen\AppData\Local\Temp\ipykernel_26400\2444976805.py:7: UserWarning: set_tic
klabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or usi
ng a FixedLocator.
  axes[i].set_xticklabels(axes[i].get_xticklabels(), rotation=90)
```



For the international plan, a higher proportion of customers who subscribed to the plan churned (42.4%) compared to those who did not subscribe (11.5%). This suggests that subscribing to the international plan may be associated with a higher likelihood of churning. This could indicate that there are issues associated with the international plans that could be making the plan unsatisfactory to the client.

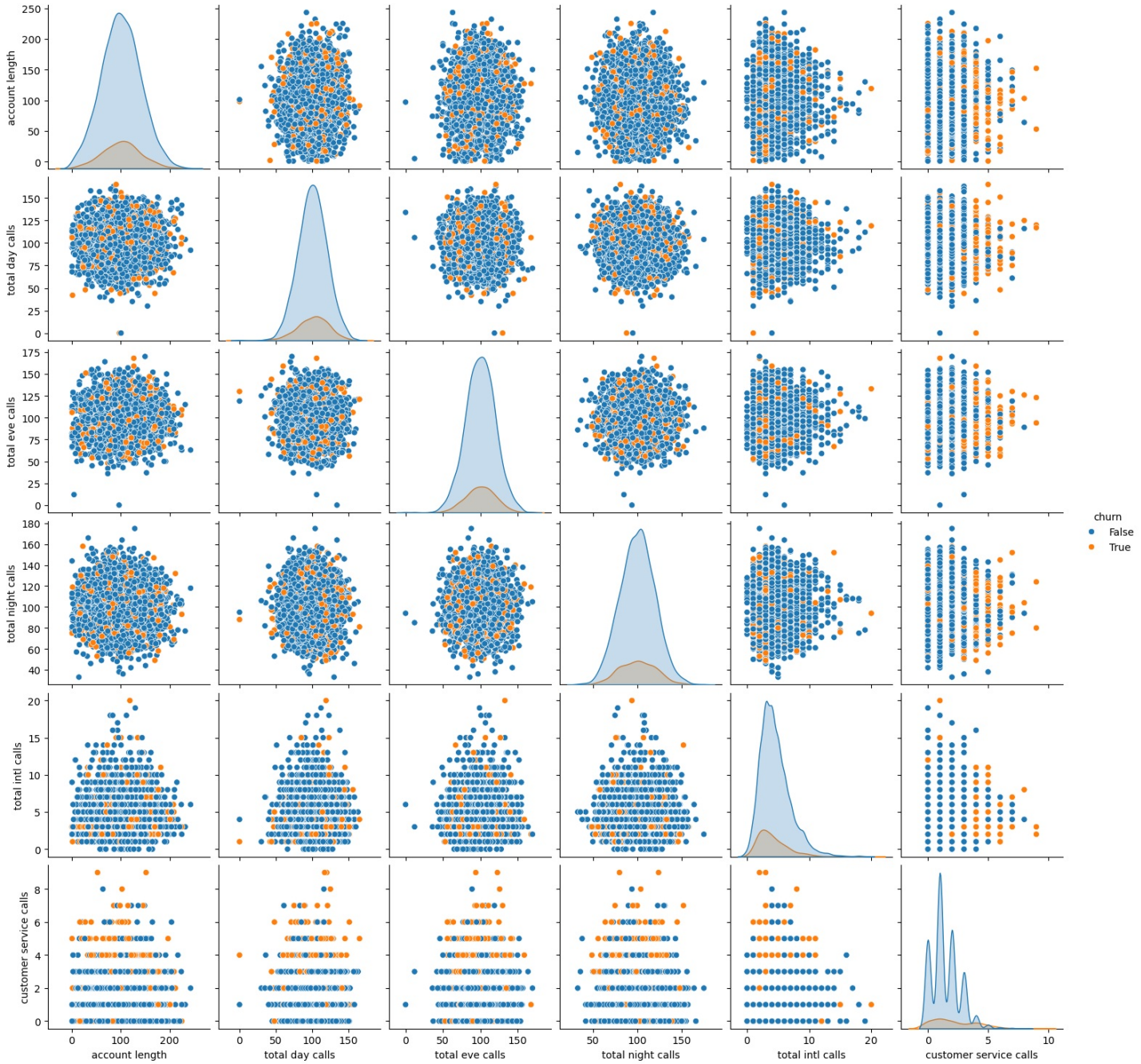
For the voice mail plan, a lower proportion of customers who subscribed to the plan churned (8.7%) compared to those who did not subscribe (16.7%). This suggests that subscribing to the voice mail plan may be associated with a lower likelihood of churning.

Correlations between different features and the target

Visualize the correlations between different features and customer churning. Here, I am trying to understand how each feature might be contributing to customer churning. I use pairplots for this case.

In [19]:

```
#plotting pairplots for numeric variables
data_temp = customer_data[["account length","total day calls","total eve calls","total ni
ght calls",
                "total intl calls","customer service calls","churn"]]
sns.pairplot(data_temp, hue="churn",height=2.5);
plt.show();
```



There seems to be strong relationship between customer service calls and true churn values. After 4 calls, customers are a lot more likely to discontinue their service.

Besides, most customer calls are are associated with disatisfaction with customer service. At this point more than 4 customer calls indicate that it takes long for their issues to be addressed, and thus a possibility of them leaving increases.

Multi-variate Analysis

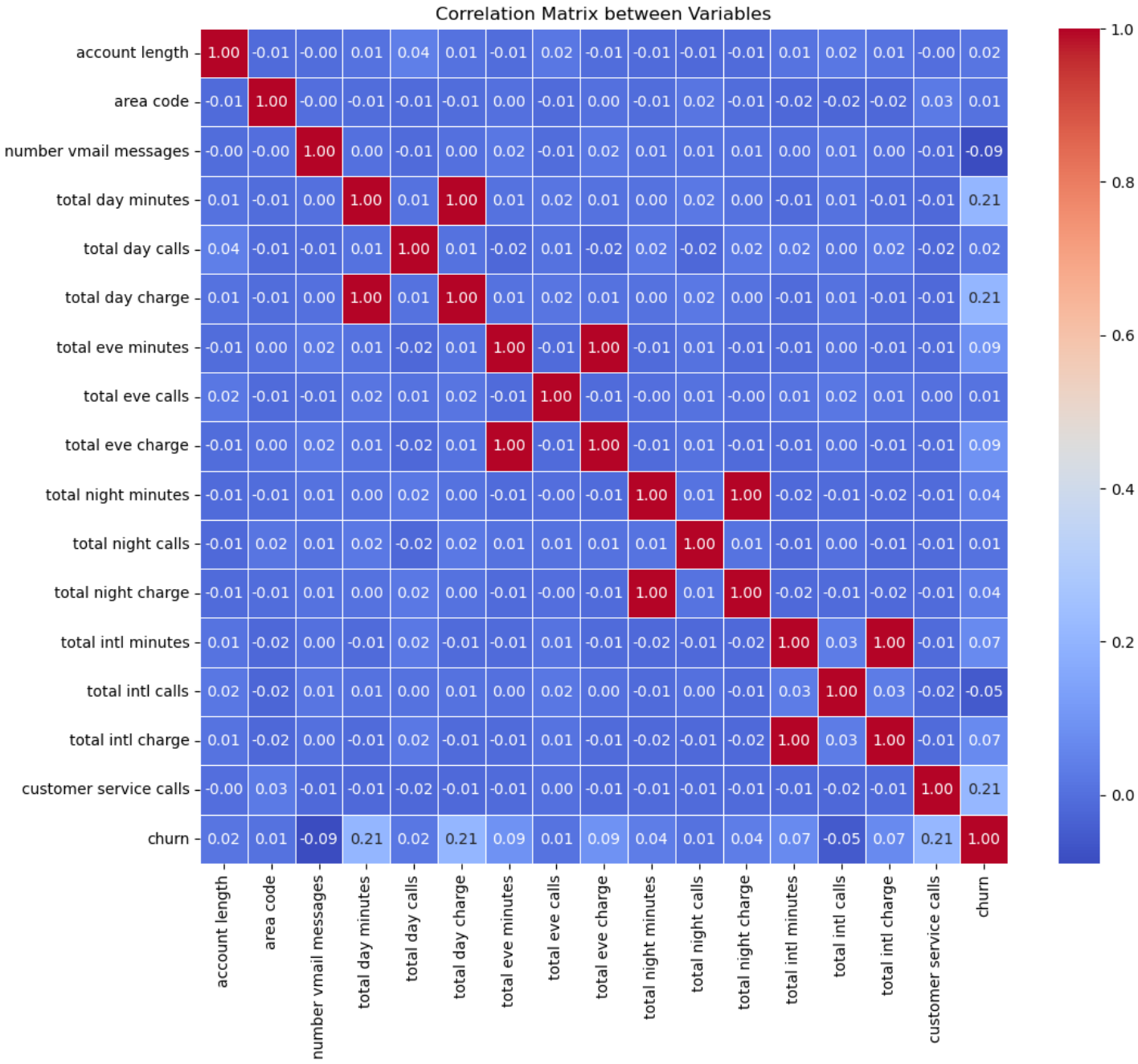
Multivariate analysis involves analyzing the relationship between multiple variables simultaneously. In this case, I explore the relationship between multiple features and the target variable (customer churn) to understand how

explore the relationship between multiple features and the target variable (customer churn) to understand how they are related when considered together.

I use a correlation matrix to identify the correlation between different variables in the dataset.

In [20]:

```
# Calculate the correlation matrix
corr_matrix = customer_data.corr(numeric_only=True)

# Generate the correlation heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Correlation Matrix between Variables')
plt.show();
```



- **There is a very low correlation between most features.**
- **However, there is a perfect positive correlation between total evening charge and total evening minutes, total day charge and total day minutes, total night charge and total night minutes, and total international charge and total international minutes. This is expected since the charge of a call depends on the length of the call in minutes. One correlated variable will have to be dropped from each pair to handle multicollinearity.**
- **total day minutes, total day charge and customer service calls have a weak positive correlation with churn.**
- **The other features have a negligible correlation with churn, approximately 0.**

Data preparation for Machine Learning

The dataset contains both categorical and numerical features, and some columns must be transformed into a format that can be used effectively by machine learning models during the modeling stage.

To achieve this, one-hot encoding will be applied to categorical columns with object data types, converting them into numerical binary variables (0s and 1s).

The data will then be divided into training and testing sets. The training set will be used to fit the models, while the test set will be used to evaluate their performance. In addition, cross-validation will be applied to ensure the models generalize well and reduce the risk of overfitting.

Finally, features that contribute little or no predictive value will be removed using regularization techniques such as Ridge and Lasso regression, which help identify and reduce the influence of irrelevant variables.

Remove columns with multicollinearity

Multicollinearity makes it hard for the model to determine the true effect of each predictor because they "move together".

In [21]:

```
# Multicollinearity can lead to issues in model performance, especially for linear models
.
# Check pairs with collinearity > 0.75
df = customer_data.corr(numeric_only=True).abs().stack().reset_index().sort_values(0, as
cending=False)
df['pairs'] = list(zip(df.level_0, df.level_1))
df.set_index(['pairs'], inplace = True)
df.drop(columns=['level_1', 'level_0'], inplace = True)
df.columns = ['cc']
df.drop_duplicates(inplace=True)
df[(df.cc>.75) & (df.cc<1)]
```

Out[21]:

pairs	cc
(total day minutes, total day charge)	1.000000
(total eve charge, total eve minutes)	1.000000
(total night charge, total night minutes)	0.999999
(total intl minutes, total intl charge)	0.999993

In [22]:

```
# Remove highly correlated features
customer_data = customer_data.drop(columns = ['total day charge', 'total eve charge', 'to
tal night charge', 'total intl charge'])
customer_data.columns
```

Out[22]:

```
Index(['state', 'account length', 'area code', 'phone number',
       'international plan', 'voice mail plan', 'number vmail messages',
       'total day minutes', 'total day calls', 'total eve minutes',
       'total eve calls', 'total night minutes', 'total night calls',
       'total intl minutes', 'total intl calls', 'customer service calls',
       'churn'],
      dtype='object')
```

In [23]:

```
# Drop the 'phone number' column as it is not relevant for prediction
```

```
customer_data = customer_data.drop(columns=['phone number'])
```

In [24]:

```
#select predictor and target variables
y = customer_data['churn']
X = customer_data.drop(columns=['churn'])
```

In [25]:

```
# split the data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=1
00)
```

In [26]:

```
# Check the X_train and y_train shapes to confirm the split
print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")
```

```
X_train shape: (2666, 15)
y_train shape: (2666,)
```

In [27]:

```
# Preview X_train.
X_train.head()
```

Out[27]:

	state	account length	area code	international plan	voice mail plan	number vmail messages	total day minutes	total day calls	total eve minutes	total eve calls	total night minutes	total night calls	total intl minutes	total intl calls	cus s
3183	KY	95	510	no	no	0	157.3	116	197.5	77	128.2	111	8.4	4	
2131	OH	52	408	no	yes	31	142.1	77	193.0	97	253.4	88	11.0	4	
1636	MN	65	510	no	yes	29	158.1	104	322.2	81	210.0	96	8.9	6	
3271	WI	62	415	no	no	0	86.3	84	238.7	99	238.4	79	12.5	1	
2295	OR	190	415	no	no	0	142.9	96	177.9	96	113.3	117	6.6	4	

Transform Categorical variables

In [28]:

```
# one-hot encode the categorical variables
X_train = pd.get_dummies(X_train, drop_first=True)
X_test = pd.get_dummies(X_test, drop_first=True)
```

In [29]:

```
# Check the encoded feature names to confirm the one-hot encoding
X_train.head()
```

Out[29]:

	account length	area code	number vmail messages	total day minutes	total day calls	total eve minutes	total eve calls	total night minutes	total night calls	total intl minutes	...	state_TX	state_UT	state_VA	sta
3183	95	510	0	157.3	116	197.5	77	128.2	111	8.4	...	False	False	False	
2131	52	408	31	142.1	77	193.0	97	253.4	88	11.0	...	False	False	False	
1636	65	510	29	158.1	104	322.2	81	210.0	96	8.9	...	False	False	False	
3271	62	415	0	86.3	84	238.7	99	238.4	79	12.5	...	False	False	False	
2295	190	415	0	142.9	96	177.9	96	113.3	117	6.6	...	False	False	False	

5 rows × 64 columns

account length	area code	number vmail messages	total day minutes	total day calls	total eve minutes	total eve calls	total night minutes	total night calls	total intl minutes	...	state_TX	state_UT	state_VA	sta

SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) is a popular technique used to address class imbalance in datasets. It works by generating synthetic samples for the minority class, which helps to balance the class distribution and improve the performance of machine learning models.

In [30]:

```
#SMOTE
from imblearn.over_sampling import SMOTE
smote = SMOTE(random_state=100)
X_train_resampled, y_train_resampled = smote.fit_resample(X_train, y_train)
```

```
c:\Users\Stephen\anaconda3\envs\learn-env\Lib\site-packages\threadpoolctl.py:1226: Runtim
eWarning:
Found Intel OpenMP ('libiomp') and LLVM OpenMP ('libomp') loaded at
the same time. Both libraries are known to be incompatible and this
can cause random crashes or deadlocks on Linux when loaded in the
same Python program.
Using threadpoolctl may cause crashes or deadlocks. For more
information and possible workarounds, please see
    https://github.com/joblib/threadpoolctl/blob/master/multiple_openmp.md

  warnings.warn(msg, RuntimeWarning)
```

In [31]:

```
# Preview synthetic sample class distribution
print(pd.Series(y_train_resampled).value_counts())
```

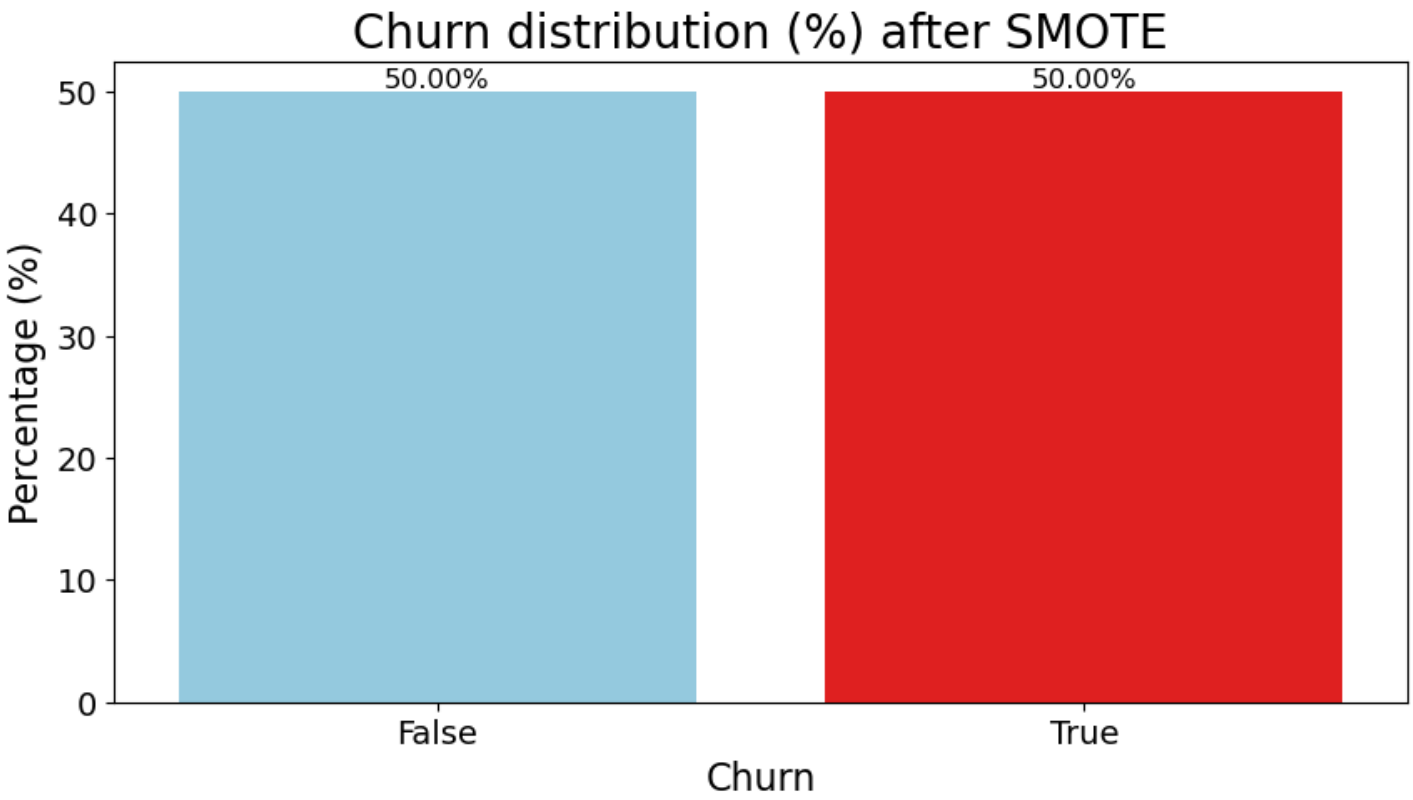
```
churn
False    2257
True     2257
Name: count, dtype: int64
```

In [32]:

```
fig, ax = plt.subplots(figsize=(10, 5))

# Get percentage distribution
churn_percent = y_train_resampled.value_counts(normalize=True).reset_index()
churn_percent.columns = ['Churn', 'Percentage']
churn_percent['Percentage'] = churn_percent['Percentage'] * 100

sns.barplot(
    data=churn_percent,
    x='Churn',
    y='Percentage',
    palette=['skyblue', 'red'],
    ax=ax
)

ax.set_title('Churn distribution (%) after SMOTE', fontsize=20)
ax.set_xlabel('Churn', fontsize=16)
ax.set_ylabel('Percentage (%)', fontsize=16)
ax.tick_params(axis='both', labelsize=14)

# Add value labels on top of bars
for p in ax.patches:
    ax.annotate(
        f'{p.get_height():.2f}%',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center',
        va='bottom',
        fontsize=12
```

```
    )

plt.show()
```

```
C:\Users\Stephen\AppData\Local\Temp\ipykernel_26400\1009675159.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. A
ssign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(
```



The distribution of the churn classes is now balanced. SMOTE was applied on the training sets only. This ensured that an accurate gauge can be made on the model's performance by using a raw test sample that has not been oversampled or undersampled.

Modeling

1. Logistic regression

Logistic regression model will be used as the baseline model.

In [33]:

```
# create a pipeline
from sklearn.pipeline import Pipeline
pipe_log = Pipeline(steps=[('scale', StandardScaler()), ('logreg', LogisticRegression(fit
_intercept=False, solver='liblinear'))])
pipe_log.fit(X_train_resampled, y_train_resampled)
```

Out[33]:

```
▸ Pipeline
    i ?
  ▸ StandardScaler
      ?
  ▸ LogisticRegression
      ?
```

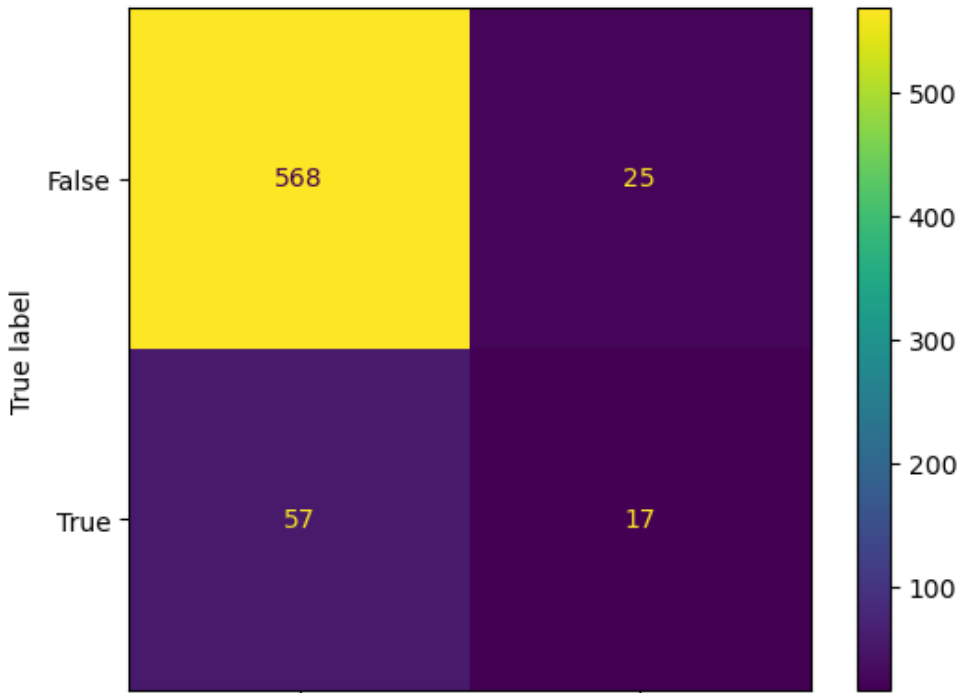
In [34]:

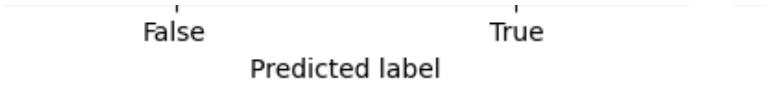
```
# function to evaluate the performance of the model
def evaluate(model, X_test, y_test):
    y_train_pred = model.predict(X_train_resampled)
    y_test_pred = model.predict(X_test)
    print('Recall_score: ')
    print('Train: ', recall_score(y_train_resampled, y_train_pred))
    print('Test: ', recall_score(y_test, y_test_pred))
    print('\nPrecision_score: ')
    print('Train: ', precision_score(y_train_resampled, y_train_pred))
    print('Test: ', precision_score(y_test, y_test_pred))
    print('\nAccuracy_score: ')
    print('Train: ', accuracy_score(y_train_resampled, y_train_pred))
    print('Test: ', accuracy_score(y_test, y_test_pred))
    print('\nClassification Report:')
    print(classification_report(y_test, y_test_pred))
    cm = confusion_matrix(y_test, y_test_pred, labels=model.classes_)
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=model.classes_)
    disp.plot();
```

In [35]:

```
# evaluate the performance of the model
evaluate(pipe_log, X_test, y_test)
```

```
Recall_score:
Train:  0.849357554275587
Test:  0.22972972972972974

Precision_score:
Train:  0.9696509863429439
Test:  0.40476190476190477

Accuracy_score:
Train:  0.9113867966326983
Test:  0.8770614692653673

Classification Report:
              precision    recall  f1-score   support

       False       0.91      0.96      0.93       593
        True       0.40      0.23      0.29        74

    accuracy                           0.88       667
   macro avg       0.66      0.59      0.61       667
weighted avg       0.85      0.88      0.86       667
```





Interpreation of the results

The model is overfitting as it performs well on the training data but does not perform well on the test data. Despite the model having a high accuracy on the test data(0.87), it has a very low recall score(0.22). The model is thus performing poorly due to the large number of false negatives. Test precision is also poor as it only predicts only 40% of the time.

Confusion Matrix interpretation

Breakdown of the values

- **True Negatives (TN) = 568 The model correctly predicted False (no churn).**
- **False Positives (FP) = 25 The model incorrectly predicted True (churn) when it was actually False.**
- **False Negatives (FN) = 57 The model missed actual churn cases (predicted False when it was True).**
- **True Positives (TP) = 17 The model correctly predicted True (churn). #### What this means**

1. **The model is very good at predicting the majority class (False) 568 correct negatives is high. This suggests the model is strong at identifying customers who do NOT churn.**
2. **Weak performance on the positive class (churn) Only 17 true positives vs 57 false negatives This means the model is missing many churners.**
3. **Recall for churn is low**

Recall (for churn) = TP / (TP + FN) = 17 / (17 + 57) ≈ 0.23

So the model only catches about 23% of actual churners, which is quite low.

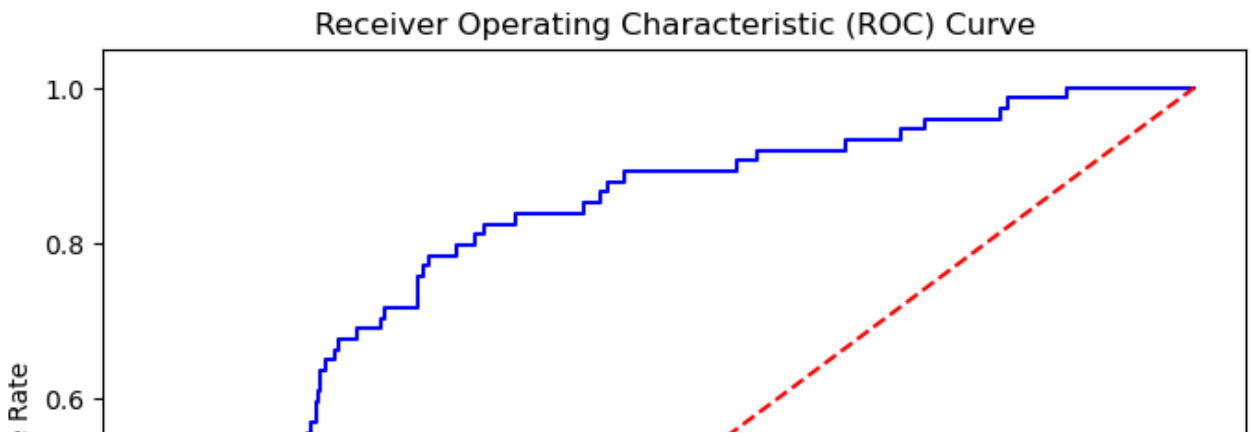
1. **Precision is moderate**

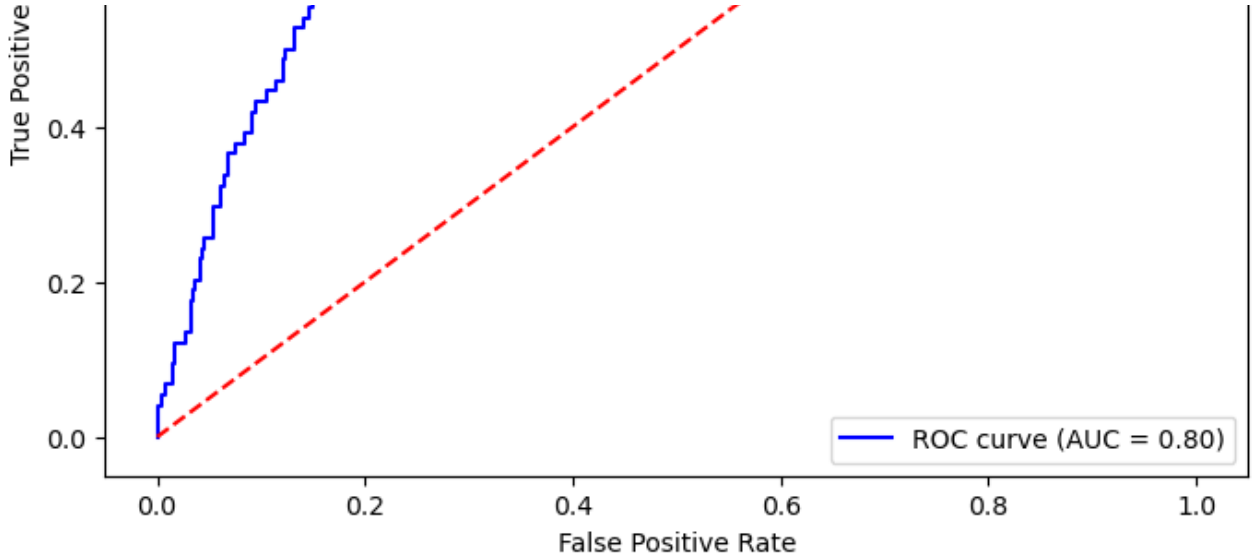
Precision = TP / (TP + FP) = 17 / (17 + 25) ≈ 0.40

So when it predicts churn, it's right about 40% of the time.

In [36]:

```
# Plot ROC curve and calculate AUC
y_test_probs = pipe_log.predict_proba(X_test)[:, 1]  # Get probabilities for the positive
class
fpr, tpr, thresholds = roc_curve(y_test, y_test_probs)
roc_auc = auc(fpr, tpr)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='blue', label=f'ROC curve (AUC = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='red', linestyle='--')  # Diagonal line for random classi
fier
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc="lower right")
plt.show()
```





Results interpretation:

- **AUC = 0.80: This is a strong indicator that the model has good predictive power. It means there is about an 80% chance the model will rank a randomly chosen positive example higher than a negative one.**
- **Curve shape (blue line): The curve is well above the diagonal red dashed line (which represents a random classifier with AUC = 0.5). This confirms the model is learning meaningful patterns rather than guessing.**
- **TPR and FPR: As you move right, the True Positive Rate (recall) increases. But the False Positive Rate also increases. The curve shows a reasonable balance—especially in the middle region where TPR improves quickly without too much increase in FPR.**

2. Decision Tree.

Decision Tree Classifier is a non-parametric supervised learning method used for classification and regression tasks.

In [37]:

```
# Create a pipeline for Decision Tree Classifier
pipe_tree = Pipeline(steps=[('scale', StandardScaler()), ('clf', DecisionTreeClassifier(
criterion='entropy', random_state=42))])
pipe_tree.fit(X_train_resampled, y_train_resampled)
```

Out[37]:

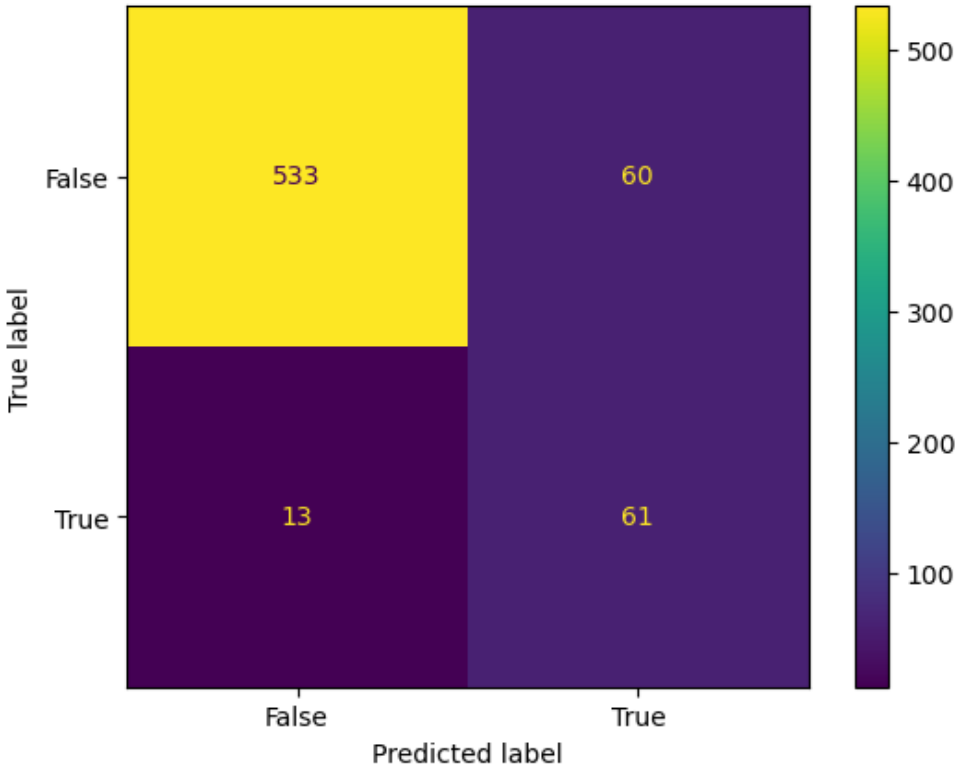
```
▸ Pipeline
                    i ?
  ▸ StandardScaler
                    ?
  ▸ DecisionTreeClassifier
                    ?
```

In [38]:

```
# Evaluate the Decision Tree model
evaluate(pipe_tree, X_test, y_test)
```

```
Recall_score:
Train:  1.0
Test:  0.8243243243243243

Precision_score:
Train:  1.0
Test:  0.5041322314049587
```

```
Accuracy_score:
Train:  1.0
Test:  0.8905547226386806

Classification Report:
              precision    recall  f1-score   support

       False       0.98      0.90      0.94       593
        True       0.50      0.82      0.63        74

    accuracy                           0.89       667
   macro avg       0.74      0.86      0.78       667
weighted avg       0.92      0.89      0.90       667
```

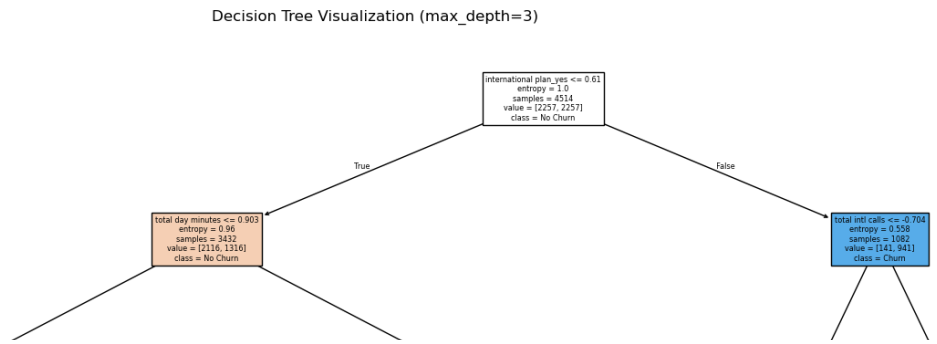


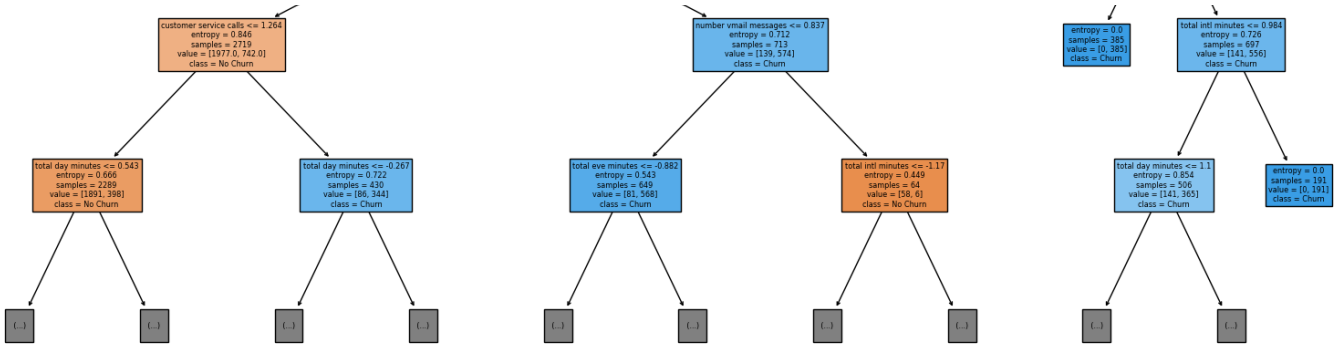
Results interpretation

- **The model performs perfectly on training data at 100% but slightly drops on test data at 86%. This suggests some overfitting.**
- **The model is very good at finding customers who will churn (high recall).**
- **But it also flags some customers incorrectly as churners (lower precision).**

In [39]:

```
# Visualize the Decision Tree
from sklearn.tree import plot_tree
plt.figure(figsize=(20, 10))
plot_tree(pipe_tree.named_steps['clf'], filled=True, feature_names=X_train_resampled.colu
mns, class_names=['No Churn', 'Churn'], max_depth=3)
plt.title('Decision Tree Visualization (max_depth=3)')
plt.show()
```





Feature importances.

In [40]:

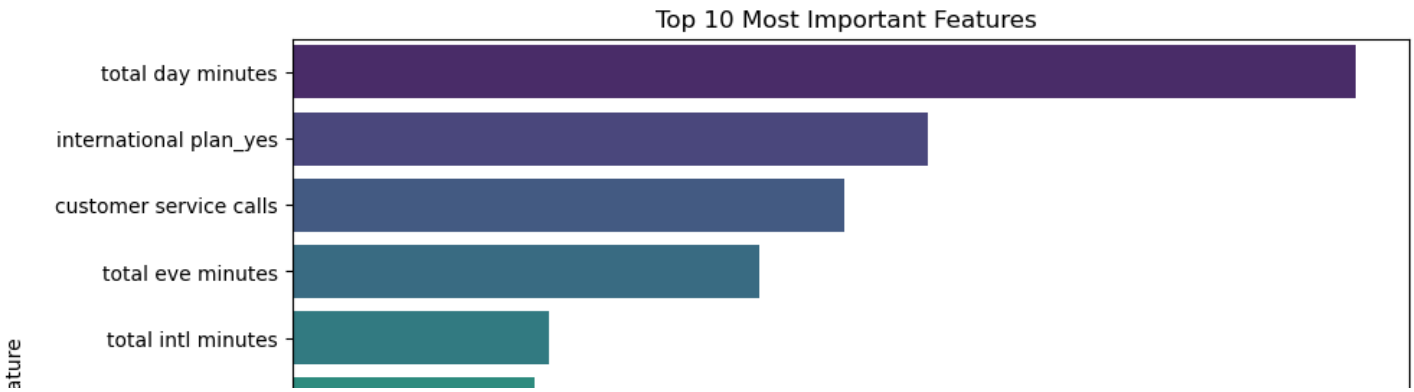
```
# Feature importance from the Decision Tree model
feature_importances = pipe_tree.named_steps['clf'].feature_importances_
feature_names = X_train_resampled.columns
feature_importance_df = pd.DataFrame({'Feature': feature_names, 'Importance': feature_im
portances})
feature_importance_df = feature_importance_df.sort_values(by='Importance', ascending=Fal
se)
print(feature_importance_df.head(10))
```

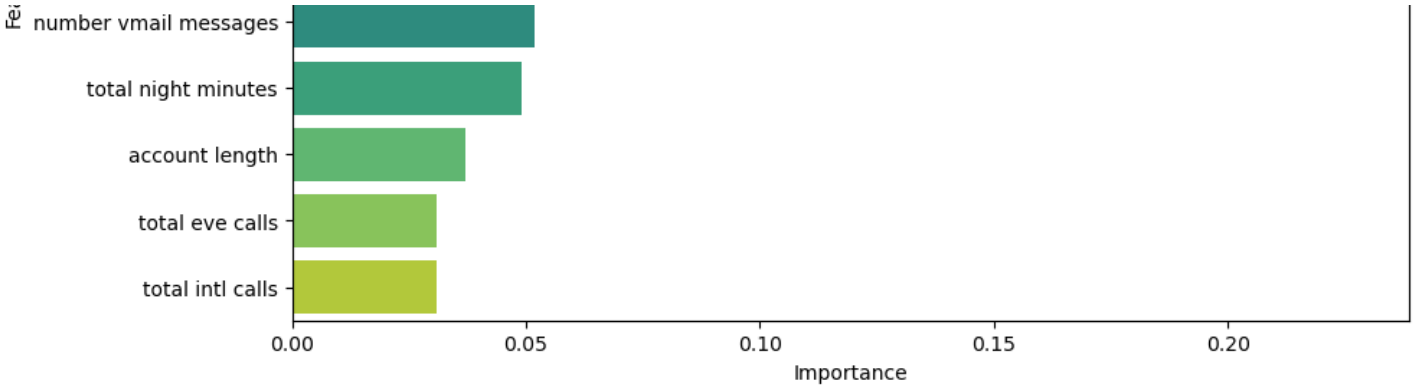
```
                   Feature  Importance
3        total day minutes    0.227270
62  international plan_yes    0.135951
11  customer service calls    0.117901
5        total eve minutes    0.099887
9       total intl minutes    0.055002
2    number vmail messages    0.051855
7      total night minutes    0.049075
0           account length    0.036929
6          total eve calls    0.030971
10        total intl calls    0.030886
```

In [ ]:

```
# Plot the top 10 most important features
plt.figure(figsize=(10, 6))
sns.barplot(x='Importance', y='Feature', data=feature_importance_df.head(10), palette='v
iridis')
plt.title('Top 10 Most Important Features')
plt.xlabel('Importance')
plt.ylabel('Feature')
plt.show()
```

```
C:\Users\Stephen\AppData\Local\Temp\ipykernel_26400\474341598.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. A
ssign the `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x='Importance', y='Feature', data=feature_importance_df.head(10), palette='
viridis')
```





total day calls, total eve calls, total night calls, and customer service calls are the most important features in predicting customer churn according to the Decision Tree model.

In [42]:

```
# Feature selection using Recursive Feature Elimination with Cross-Validation (RFECV)
rfecv = RFECV(estimator=DecisionTreeClassifier(random_state=42),scoring='recall')
pipe_tree1 = Pipeline(steps=[('scale', StandardScaler()), ('Feature Selection', rfecv),
('clf', DecisionTreeClassifier(random_state=42))])
pipe_tree1.fit(X_train_resampled, y_train_resampled)
```

Out[42]:

```
▸ Pipeline
                                    i ?
  ▸ StandardScaler
                                    ?
  ▸ Feature Selection: RFECV
                                    ?
  ▸ estimator:
    DecisionTreeClassifier
    ▸ DecisionTreeClassifier
                                    ?
  ▸ DecisionTreeClassifier
                                    ?
```

In [43]:

```
# optimal number of features selected via cross-validation
print(f'Optimal number of features: {rfecv.n_features_}' )
```

```
Optimal number of features: 55
```

In [44]:

```
# selected features (assigned a rank of 1)
rfecv_df = pd.DataFrame(rfecv.ranking_,index=X_train_resampled.columns,columns=['Rank'])
.sort_values(by='Rank',ascending=True)
rfecv_df[rfecv_df['Rank'] == 1]
```

Out[44]:

	Rank
account length	1
area code	1
number vmail messages	1
total day minutes	1
total day calls	1

	Rank
total eve minutes	1
total eve calls	1
total night minutes	1
total night calls	1
total intl minutes	1
total intl calls	1
customer service calls	1
state_AL	1
state_AZ	1
state_CA	1
state_CO	1
state_LA	1
state_DC	1
state_GA	1
state_FL	1
state_KS	1
state_IN	1
state_IL	1
state_ID	1
state_MD	1
state_MA	1
state_MT	1
state_ND	1
state_MS	1
state_MN	1
state_MI	1
state_ME	1
state_WV	1
state_VT	1
state_SC	1
state_SD	1
state_TX	1
state_TN	1
state_NH	1
state_NE	1
state_NV	1
state_NY	1
state_NJ	1
state_NM	1
state_OH	1
state_OK	1
state_PA	1
state_OR	1
international plan_yes	1
state_WY	1

	Rank
state_WA	1
state_WI	1
voice mail plan_yes	1
state_VA	1
state_UT	1

In [45]:

```
# remove columns that are not part of the optimal features
cols = rfecv_df[rfecv_df['Rank'] == 1].index
X_train_resampled = X_train_resampled[cols]
X_test = X_test[cols]
X_train_resampled.head(2)
```

Out[45]:

	account length	area code	number vmail messages	total day minutes	total day calls	total eve minutes	total eve calls	total night minutes	total night calls	total intl minutes	...	state_OK	state_PA	state_OR	intern p
0	95	510	0	157.3	116	197.5	77	128.2	111	8.4	...	False	False	False	
1	52	408	31	142.1	77	193.0	97	253.4	88	11.0	...	False	False	False	

2 rows × 55 columns

Hyperparameter tuning of the Decision tree

In [46]:

```
# hyperparameter tuning using GridSearchCV
params_dt = {'clf__criterion': ['gini', 'entropy'],
             'clf__max_depth': range(14, 32, 2),
             'clf__min_samples_split' : range(2, 10, 2),
             'clf__min_samples_leaf': [2, 3, 5, 7, 10],
             'clf__max_features': [11, 13, 15]
}

gridsearch_dt = GridSearchCV(pipe_tree, params_dt, cv=4, scoring='recall')
gridsearch_dt.fit(X_train_resampled, y_train_resampled)
```

Out[46]:

```
▸ GridSearchCV
                    i ?
▸ best_estimator_: Pipeline
  ▸ StandardScaler
                    ?
  ▸
  DecisionTreeClassifier
                    ?
```

In [47]:

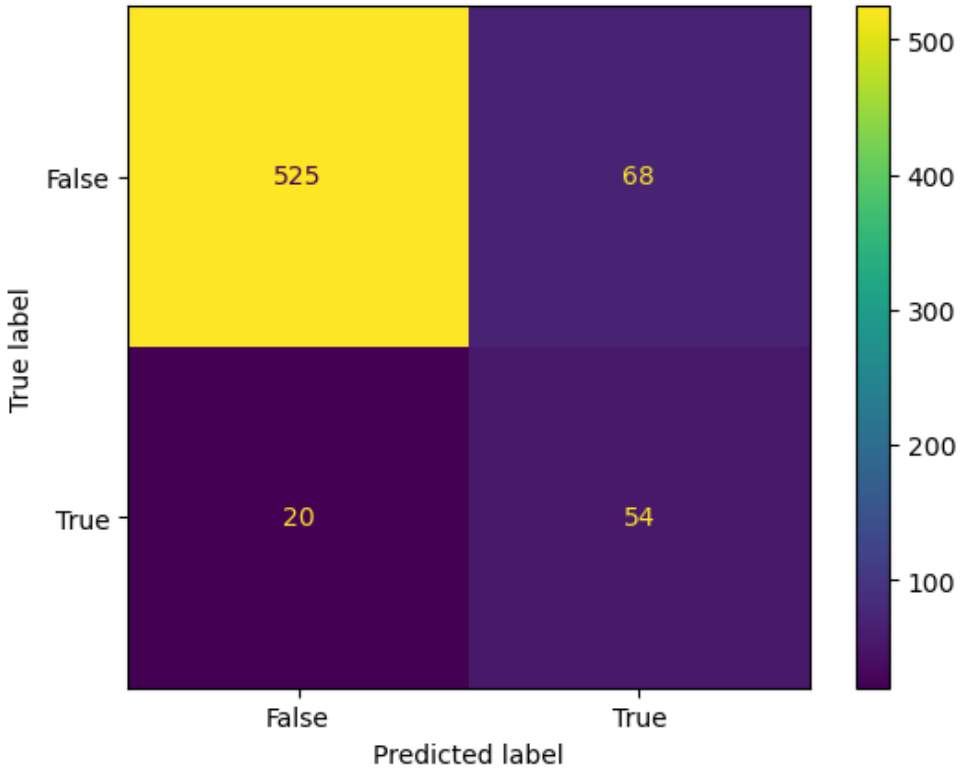
```
print(f"Best parameters: {gridsearch_dt.best_params_}")
print(f"Best recall score from GridSearchCV: {gridsearch_dt.best_score_:.4f}")
```

```
Best parameters: {'clf__criterion': 'entropy', 'clf__max_depth': 24, 'clf__max_features'
: 15, 'clf__min_samples_leaf': 3, 'clf__min_samples_split': 2}
Best recall score from GridSearchCV: 0.8994
```

In [48]:

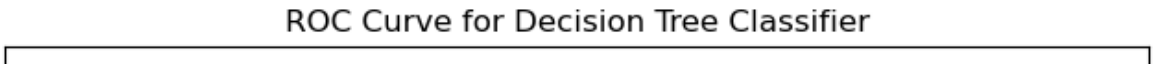
```
# Evaluate the performance of the best Decision Tree model from GridSearchCV
evaluate(gridsearch_dt, X_test, y_test)
```

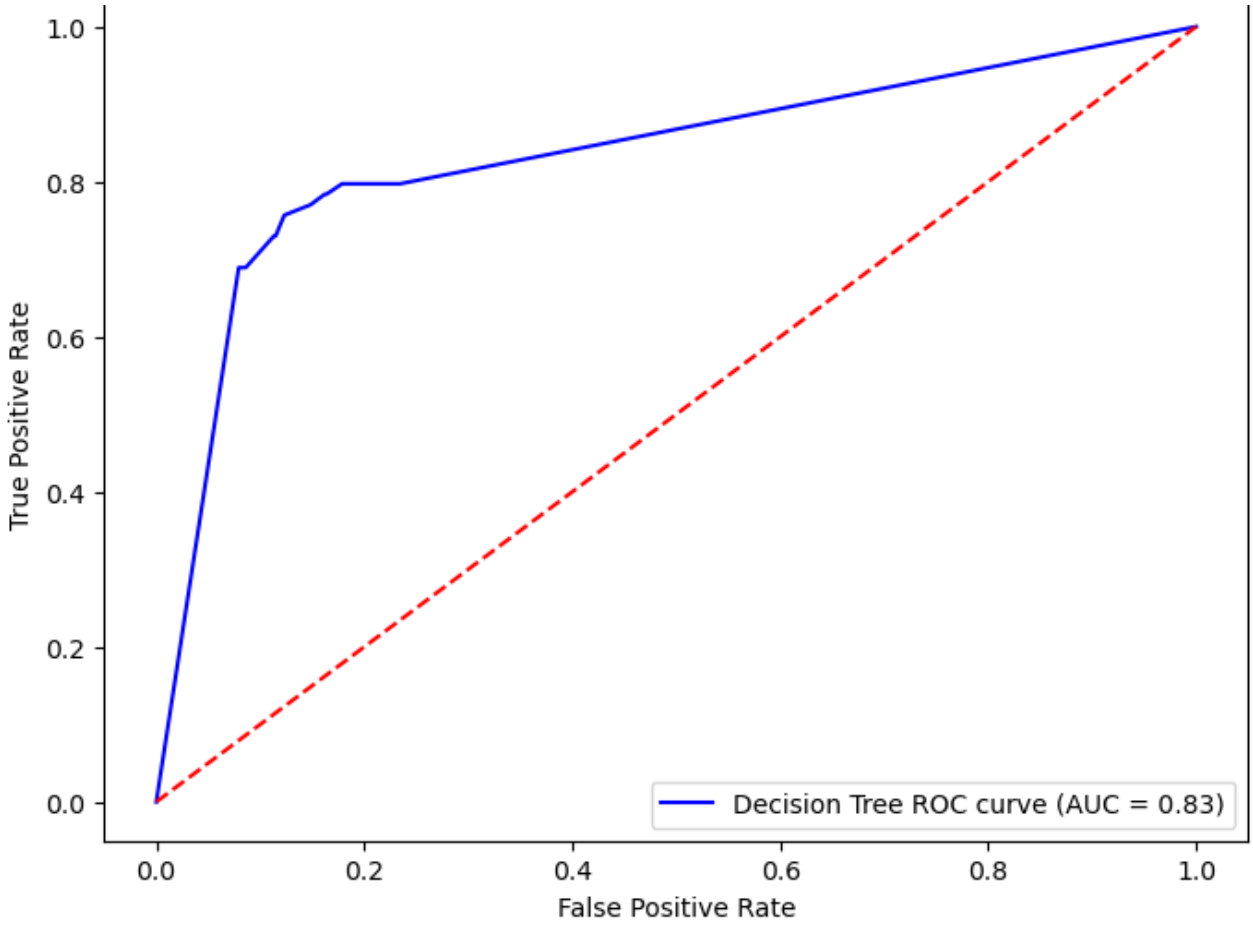
```
Recall_score:
Train:  0.9689853788214444
Test:  0.7297297297297297

Precision_score:
Train:  0.9798387096774194
Test:  0.4426229508196721

Accuracy_score:
Train:  0.9745237040319008
Test:  0.8680659670164917

Classification Report:
              precision    recall  f1-score   support

       False       0.96      0.89      0.92       593
        True       0.44      0.73      0.55        74

    accuracy                           0.87       667
   macro avg       0.70      0.81      0.74       667
weighted avg       0.91      0.87      0.88       667
```



In [49]:

```
# Plot ROC curve and calculate AUC for Decision Tree
y_test_probs_dt = gridsearch_dt.predict_proba(X_test)[:, 1]
fpr_dt, tpr_dt, thresholds_dt = roc_curve(y_test, y_test_probs_dt)
roc_auc_dt = auc(fpr_dt, tpr_dt)
plt.figure(figsize=(8, 6))
plt.plot(fpr_dt, tpr_dt, color='blue', label=f'Decision Tree ROC curve (AUC = {roc_auc_
dt:.2f})')
plt.plot([0, 1], [0, 1], color='red', linestyle='--')  # Diagonal line for random class
ifier
plt.title('ROC Curve for Decision Tree Classifier')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc="lower right")
plt.show()
```





Results interpretation

The model is performing better after hyperparameter tuning, based on the recall score of 73%. This is a strong performance for churn predictiob. However, the precision score is slightly low. There is a decrease in the difference between the train scores and the test scores(overfitting reduced).

Random Forest Classifier

A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control over-fitting.

In [50]:

```
# Random Forest Classifier
pipe_rf = Pipeline(steps=[('scale', StandardScaler()), ('rf', RandomForestClassifier(ra
ndom_state=42))])
# Fit the Random Forest model on the resampled training data
pipe_rf.fit(X_train_resampled, y_train_resampled)
```

Out[50]:

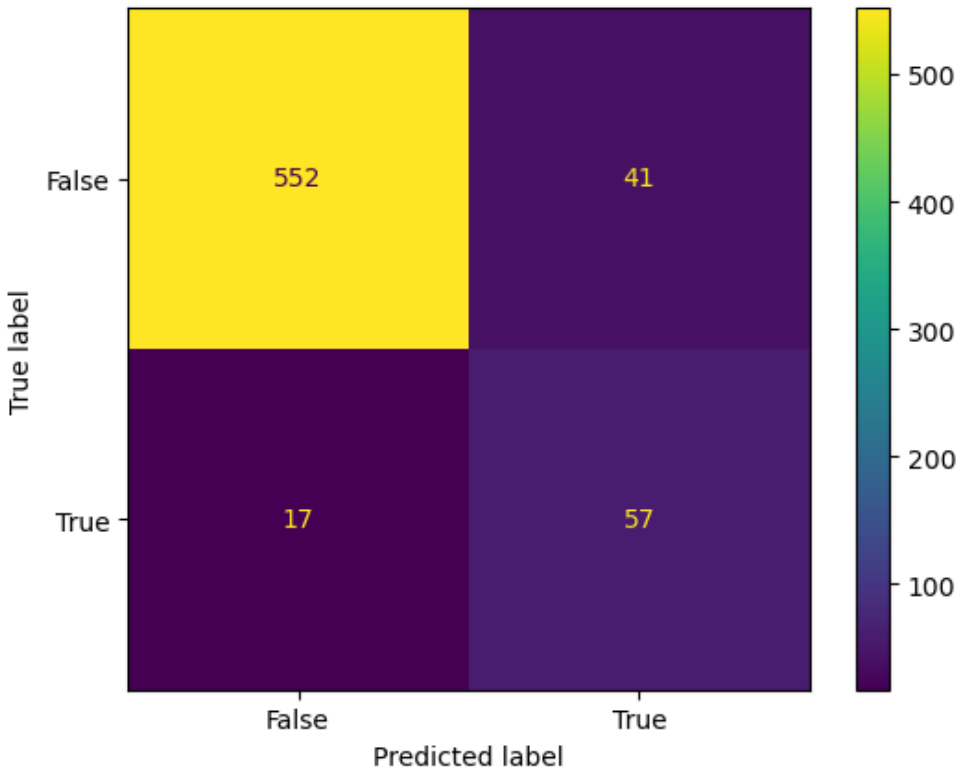
```
▸ Pipeline
    i ?
  ▸ StandardScaler
    ?
  ▸ RandomForestClassifier
    ?
```

In [51]:

```
# Evaluate the performance of the Random Forest model
evaluate(pipe_rf, X_test, y_test)
```

Recall score:

```
Recall_score:
Train:  1.0
Test:  0.7702702702702703

Precision_score:
Train:  1.0
Test:  0.5816326530612245

Accuracy_score:
Train:  1.0
Test:  0.9130434782608695

Classification Report:
              precision    recall  f1-score   support

       False       0.97      0.93      0.95       593
        True       0.58      0.77      0.66        74

    accuracy                           0.91       667
   macro avg       0.78      0.85      0.81       667
weighted avg       0.93      0.91      0.92       667
```

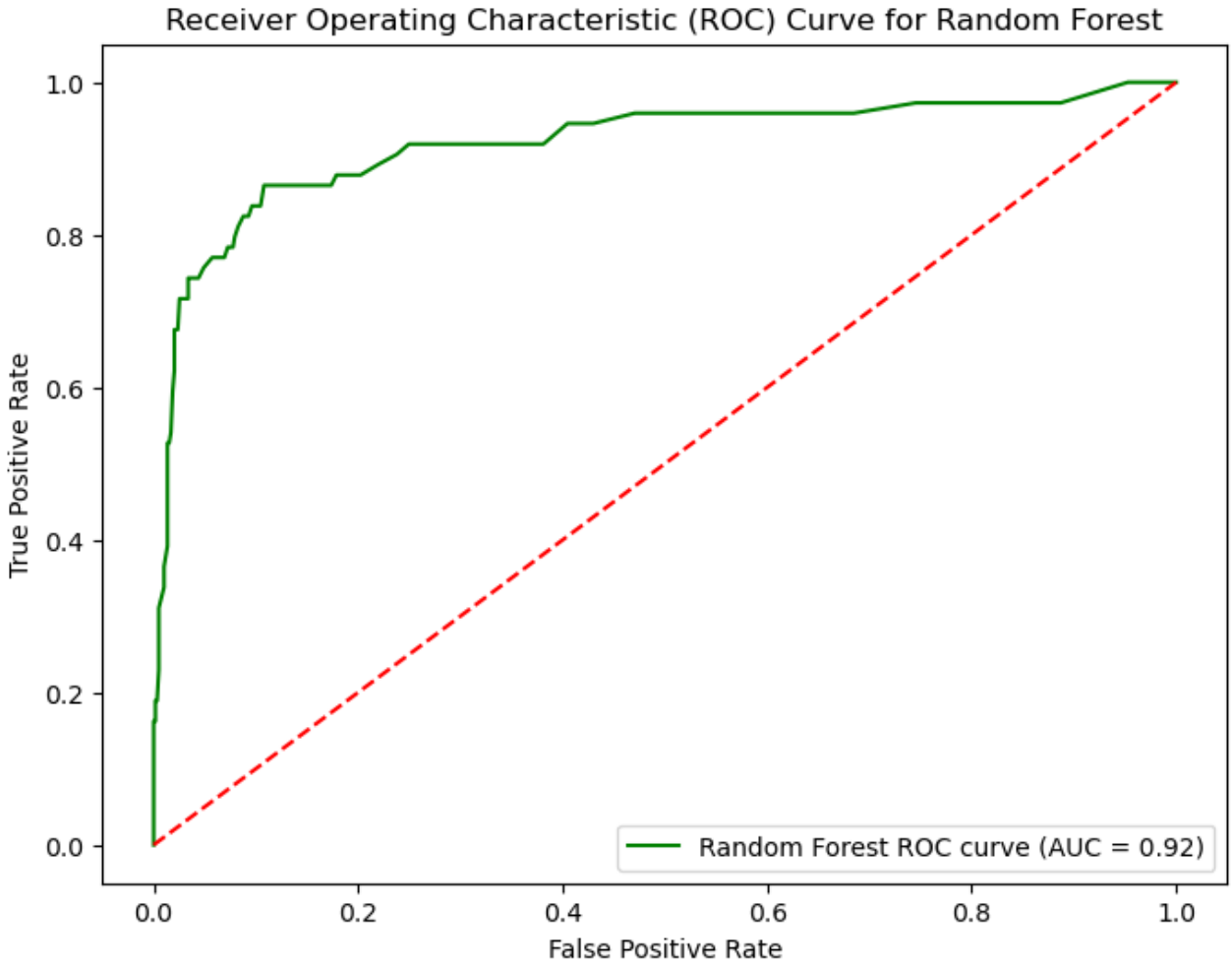


Results interpretation

The recall is 77% which is a strong churn prediction.The precicion is also better at 58%

In [52]:

```
# Plot ROC curve and calculate AUC for Random Forest
y_test_probs_rf = pipe_rf.predict_proba(X_test)[:, 1]  # Get probabilities for the posi
tive class
fpr_rf, tpr_rf, thresholds_rf = roc_curve(y_test, y_test_probs_rf)
roc_auc_rf = auc(fpr_rf, tpr_rf)
plt.figure(figsize=(8, 6))
plt.plot(fpr_rf, tpr_rf, color='green', label=f'Random Forest ROC curve (AUC = {roc_auc
_rf:.2f})')
plt.plot([0, 1], [0, 1], color='red', linestyle='--')  # Diagonal line for random class
ifier
plt.title('Receiver Operating Characteristic (ROC) Curve for Random Forest')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc="lower right")
plt.show()
```



Results interpretation

The AUC is 0.92 which is actually very good.

Hyperparameter tuning of Random forest model

Hyperparameter tuning for Random Forest using RandomizedSearchCV. We will use RandomizedSearchCV instead of GridSearchCV for Random Forest due to the larger number of hyperparameters and their possible values, which can lead to a combinatorial explosion in the number of models to train.

In [53]:

```
#  RandomizedSearchCV allows us to sample a fixed number of hyperparameter combinations
,
#  making it more efficient while still providing good results.
from sklearn.model_selection import RandomizedSearchCV
pipe_rf = Pipeline([
    ('rf', RandomForestClassifier(random_state=200))
])

params_rf = {
    'rf__n_estimators': [100, 200, 400],
    'rf__max_depth': [5, 10, 15, 20, None],
    'rf__min_samples_split': [2, 5, 10],
    'rf__min_samples_leaf': [1, 2, 5],
    'rf__max_features': ['sqrt', 'log2'],
    'rf__criterion': ['gini', 'entropy']
}

randomsearch_rf = RandomizedSearchCV(
    pipe_rf,
    params_rf,
    n_iter=30,        # only 30 combinations tested
    cv=4,
```

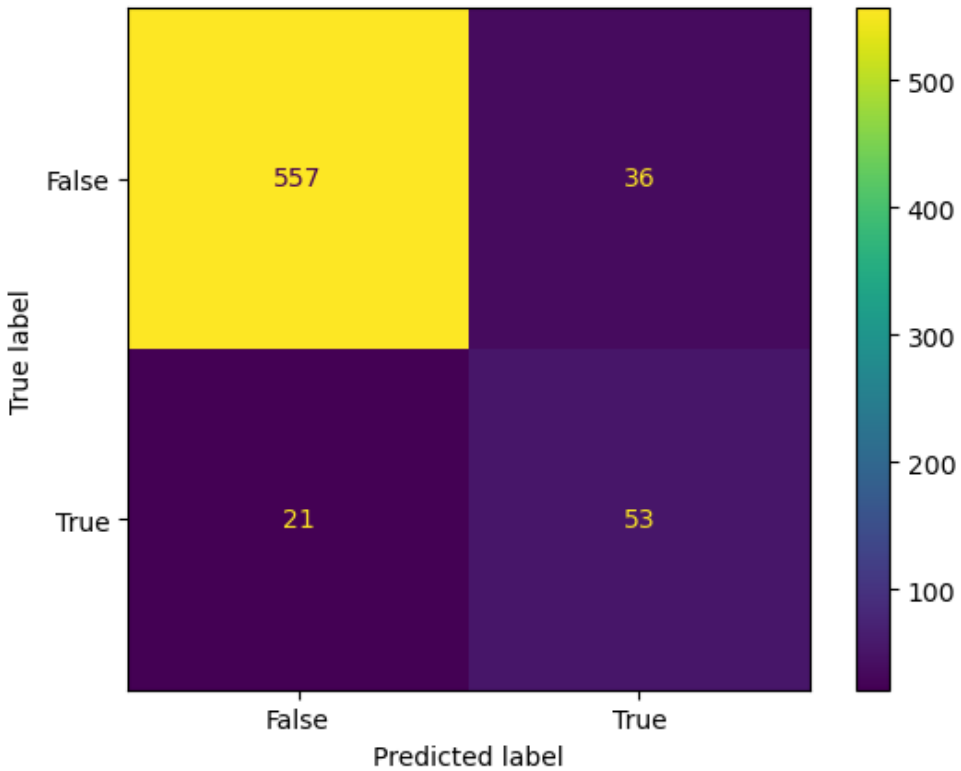
```
    scoring='recall',
    n_jobs=-1,
    random_state=200
)

randomsearch_rf.fit(X_train_resampled, y_train_resampled)

print("Best Parameters:", randomsearch_rf.best_params_)
print("Best CV Recall:", randomsearch_rf.best_score_)
```

```
Best Parameters: {'rf__n_estimators': 200, 'rf__min_samples_split': 5, 'rf__min_samples_
leaf': 1, 'rf__max_features': 'log2', 'rf__max_depth': None, 'rf__criterion': 'gini'}
Best CV Recall: 0.9521480574907426
```

In [54]:

```
# evaluate the performance of the model
evaluate(randomsearch_rf, X_test, y_test)
```

```
Recall_score:
Train:  0.9986708019494904
Test:  0.7162162162162162

Precision_score:
Train:  1.0
Test:  0.5955056179775281

Accuracy_score:
Train:  0.9993354009747453
Test:  0.9145427286356822

Classification Report:
              precision    recall  f1-score   support

       False       0.96      0.94      0.95       593
        True       0.60      0.72      0.65        74

    accuracy                           0.91       667
   macro avg       0.78      0.83      0.80       667
weighted avg       0.92      0.91      0.92       667
```



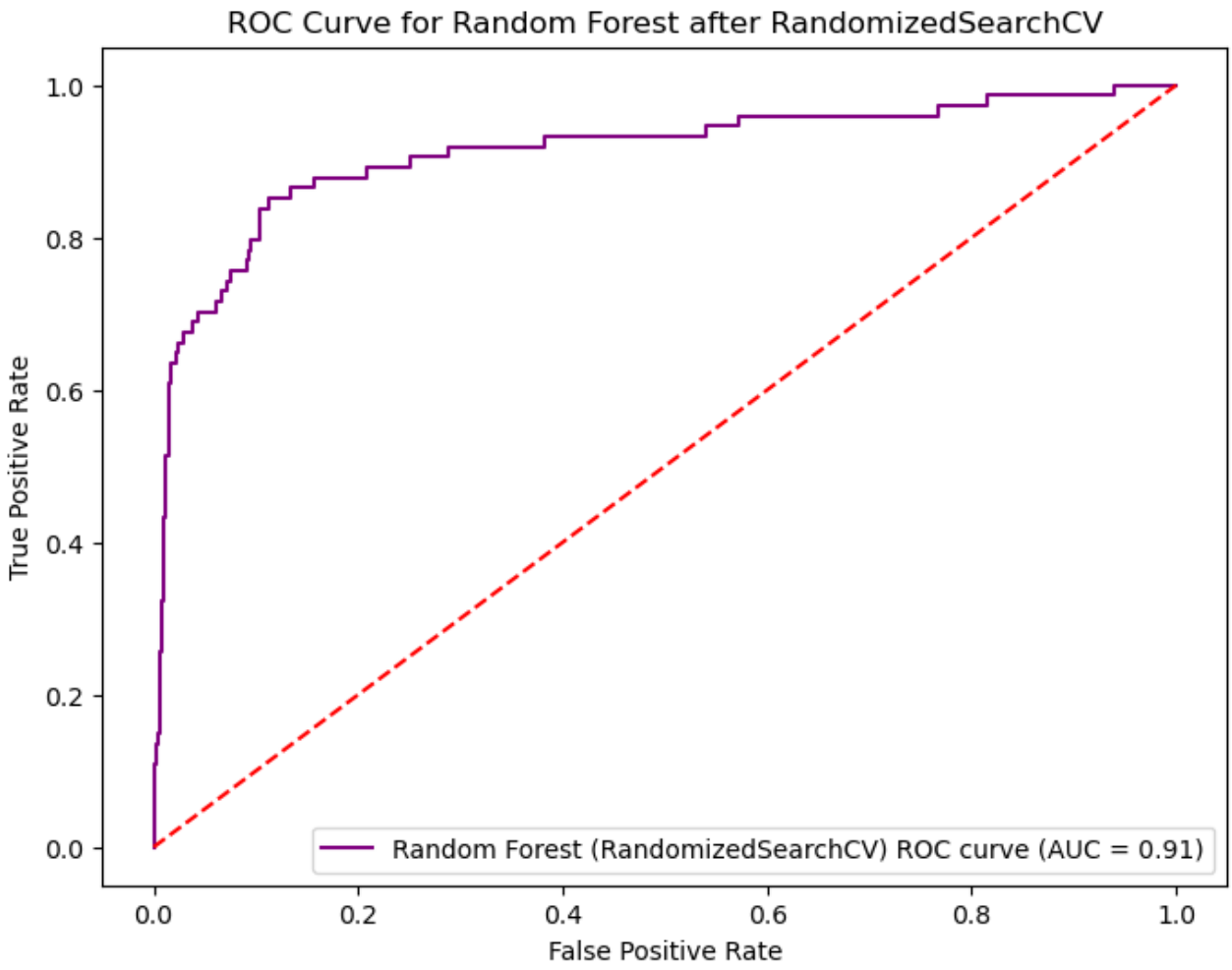
Results interpretation

The model correctly identifies about 70% of actual churners, correctly predicts 0.59% of the time and an

The model correctly identifies about 72% of actual churners, correctly predicts 0.59% of the time and an accuracy score of 92%

In [55]:

```
# Plot ROC curve and calculate AUC for Random Forest after RandomizedSearchCV
y_test_probs_rf_rs = randomsearch_rf.predict_proba(X_test)[:, 1]  # Get probabilities f
or the positive class
fpr_rf_rs, tpr_rf_rs, thresholds_rf_rs = roc_curve(y_test, y_test_probs_rf_rs)
roc_auc_rf_rs = auc(fpr_rf_rs, tpr_rf_rs)
plt.figure(figsize=(8, 6))
plt.plot(fpr_rf_rs, tpr_rf_rs, color='purple', label=f'Random Forest (RandomizedSearchC
V) ROC curve (AUC = {roc_auc_rf_rs:.2f})')
plt.plot([0, 1], [0, 1], color='red', linestyle='--')  # Diagonal line for random class
ifier
plt.title('ROC Curve for Random Forest after RandomizedSearchCV')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc="lower right")
plt.show()
```



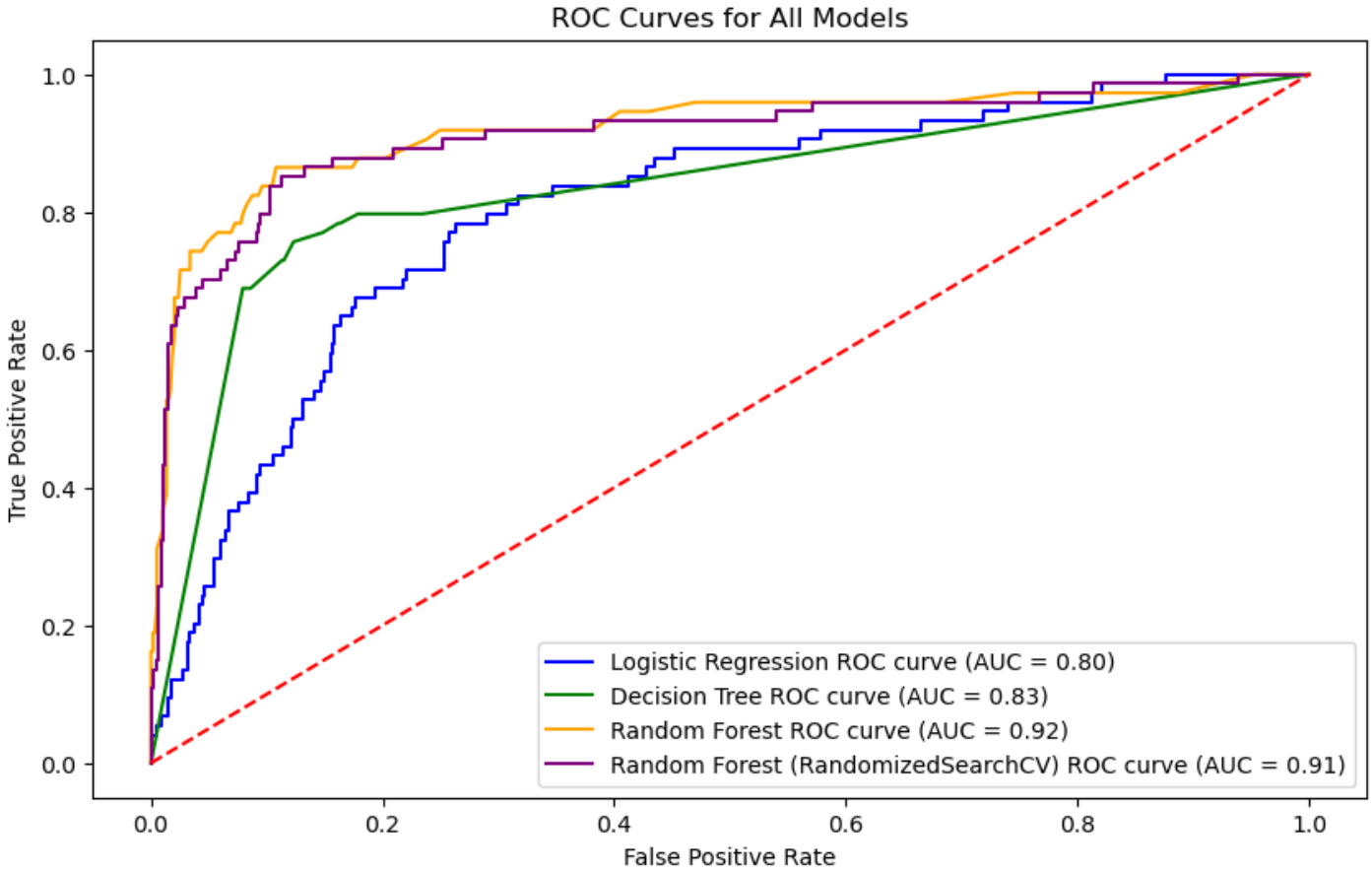
Results interpretation

AUC is 0.91 for the Random Forest model after hyperparameter tuning with RandomizedSearchCV, which indicates excellent model performance in distinguishing between the classes.

In [56]:

```
# Plot AUC and compare all models
plt.figure(figsize=(10, 6))
plt.plot(fpr, tpr, color='blue', label=f'Logistic Regression ROC curve (AUC = {roc_auc:
.2f})')
plt.plot(fpr_dt, tpr_dt, color='green', label=f'Decision Tree ROC curve (AUC = {roc_auc
_dt:.2f})')
plt.plot(fpr_rf, tpr_rf, color='orange', label=f'Random Forest ROC curve (AUC = {roc_au
c_rf:.2f})')
plt.plot(fpr_rf_rs, tpr_rf_rs, color='purple', label=f'Random Forest (RandomizedSearchC
```

```
V) ROC curve (AUC = {roc_auc_rf_rs:.2f})')
plt.plot([0, 1], [0, 1], color='red', linestyle='--')  # Diagonal line for random class
ifier
plt.title('ROC Curves for All Models')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc= 'lower right')
plt.show()
```



In [61]:

```
model_comparison = pd.DataFrame({'Model':['Logistic Regression',
                                          'Decision Trees Classifier',
                                          'Random Forest Classifier'],
                                 'Accuracy (Test Set)':[0.91,0.89,0.91],
                                 'F1 Score (Test Set)':[0.29,0.63,0.66],
                                 'Recall (Test Set)':[0.22,0.82,0.77],
                                 'Precision (Test Set)':[0.40,0.50,0.58]})

model_comparison.style.highlight_max(color = 'grey', axis = 0)
```

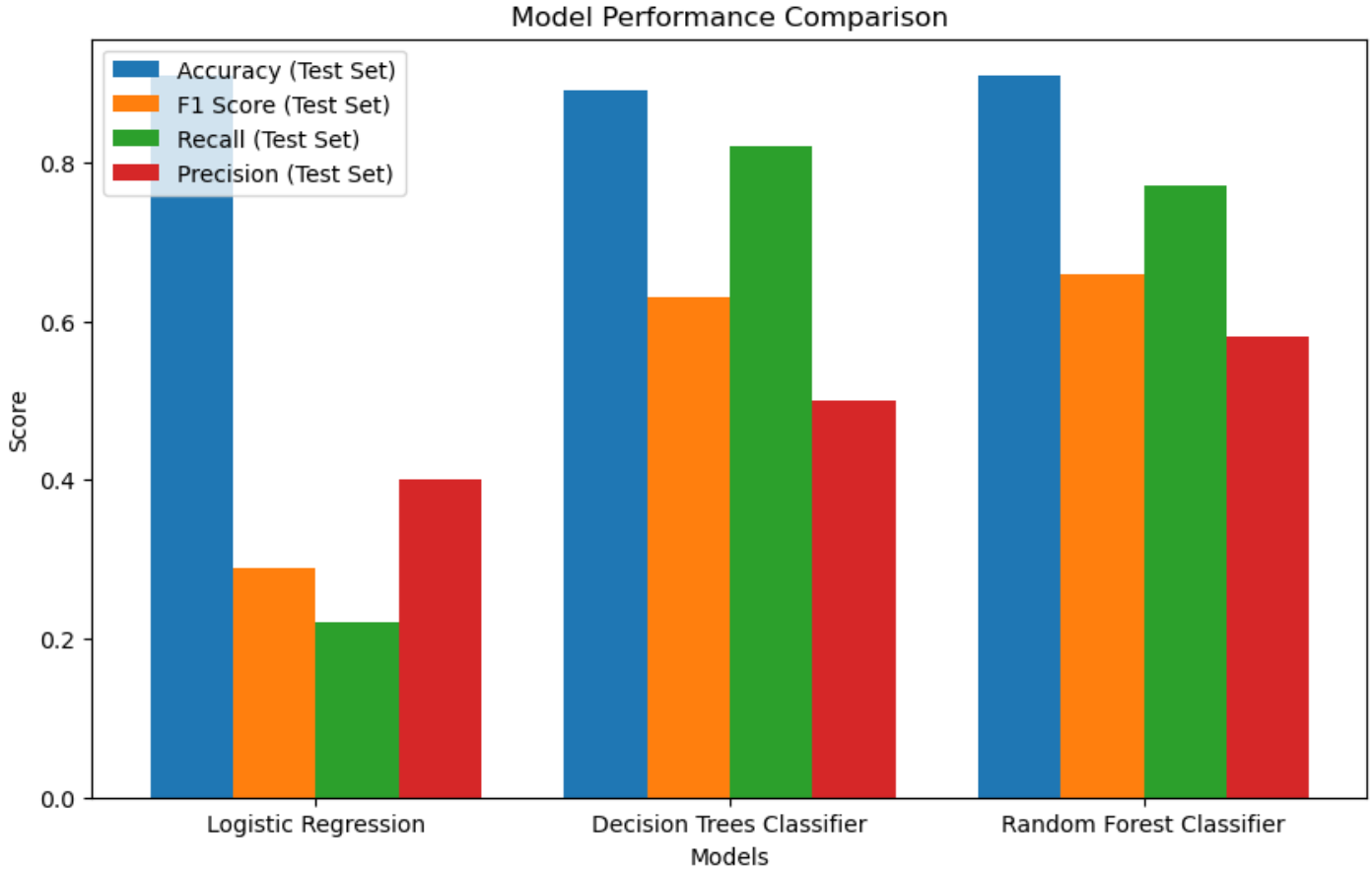
Out[61]:

	Model	Accuracy (Test Set)	F1 Score (Test Set)	Recall (Test Set)	Precision (Test Set)
0	Logistic Regression	0.910000	0.290000	0.220000	0.400000
1	Decision Trees Classifier	0.890000	0.630000	0.820000	0.500000
2	Random Forest Classifier	0.910000	0.660000	0.770000	0.580000

In [ ]:

```
# Plot a bar graph to compare the models performance
metrics = ['Accuracy (Test Set)', 'F1 Score (Test Set)', 'Recall (Test Set)', 'Precisio
n (Test Set)']
models = model_comparison['Model']

x = np.arange(len(models))
width = 0.2
```

```
fig, ax = plt.subplots(figsize=(10, 6))

for i, metric in enumerate(metrics):
    ax.bar(x + i*width, model_comparison[metric], width, label=metric)

ax.set_xlabel('Models')
ax.set_ylabel('Score')
ax.set_title('Model Performance Comparison')
ax.set_xticks(x + width*1.5)
ax.set_xticklabels(models)
ax.legend()

plt.show()
```



Conclusion

The analysis of the dataset indicates that customer churn can be effectively predicted using machine learning models. Of the models evaluated, the Random Forest Classifier is recommended as it shows the strongest overall performance. It achieves a higher ROC curve that closely approaches the top-left corner, reflecting a high AUC (Area Under the Curve) and strong ability to distinguish between churn and non-churn customers.

Based on these results, it is recommended that Syriatel adopt the Random Forest Classifier as the primary model for churn prediction. This model performs consistently well across key evaluation metrics, including accuracy, F1-score, recall, and precision on the test set, making it a reliable choice for identifying customers who are at risk of leaving.

Key features influencing customer churn include:

- **Total day minutes: total duration of daytime calls**
- **Total evening minutes: total duration of evening calls**
- **Customer service calls: number of interactions with customer support**
- **Total international minutes: total duration of international calls**

Business recommendations for Syriatel

From a strategic perspective, Syriatel should implement customer retention initiatives that target the key drivers of churn, particularly those related to call usage and pricing. This may include offering personalized

discounts or more competitive rates for daytime usage to encourage continued customer engagement and reduce attrition.

In addition, the company should focus on improving customer service efficiency and satisfaction. Since the number of customer service calls is a significant predictor of churn, reducing the need for repeated or unresolved support interactions could help improve customer experience and retention.

Overall, by addressing these key factors, Syriatel can reduce churn rates, strengthen customer loyalty, and minimize revenue loss.